

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 3

Aim: Creating Advanced Visualizations in Tableau

- a) Create stacked and grouped bar charts for comparison
- b) Build treemaps and pie charts for part-to-whole analysis
- c) Design scatter plots to identify relationships
- d) Create dual-axis and combination charts
- e) Develop histograms for distribution analysis
- f) Build box-and-whisker plots
- g) Use Gantt charts and advanced date visualizations

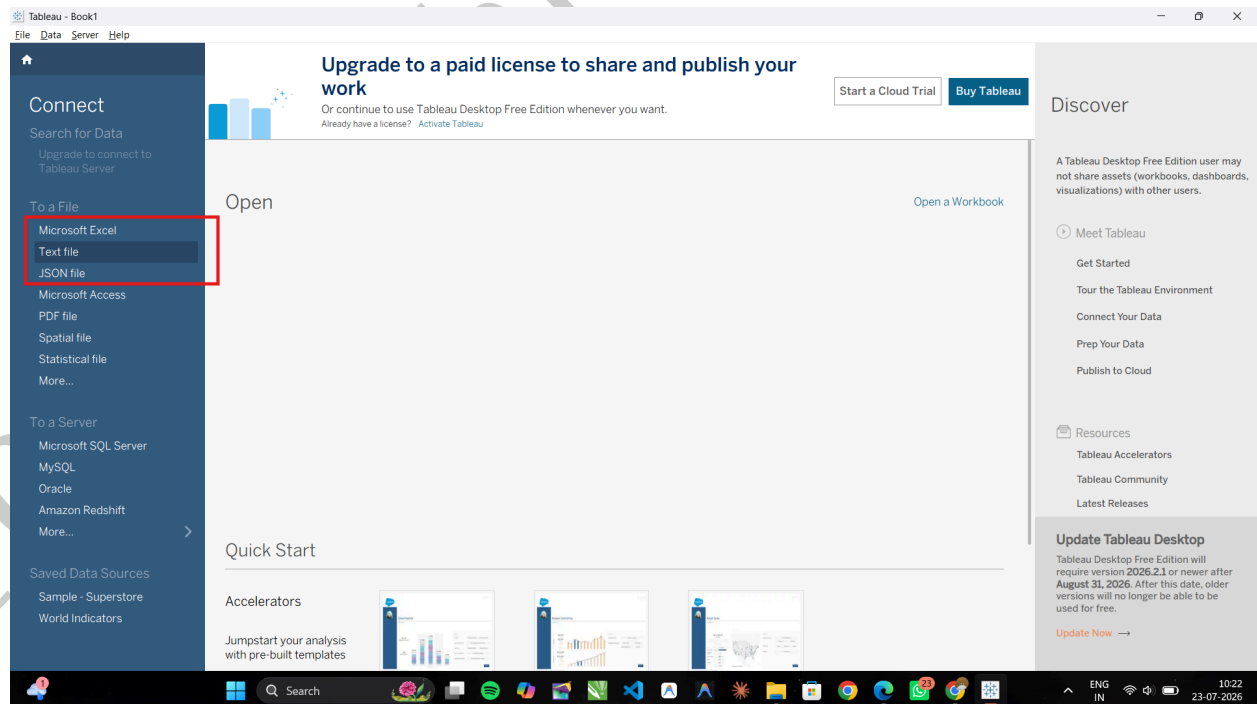
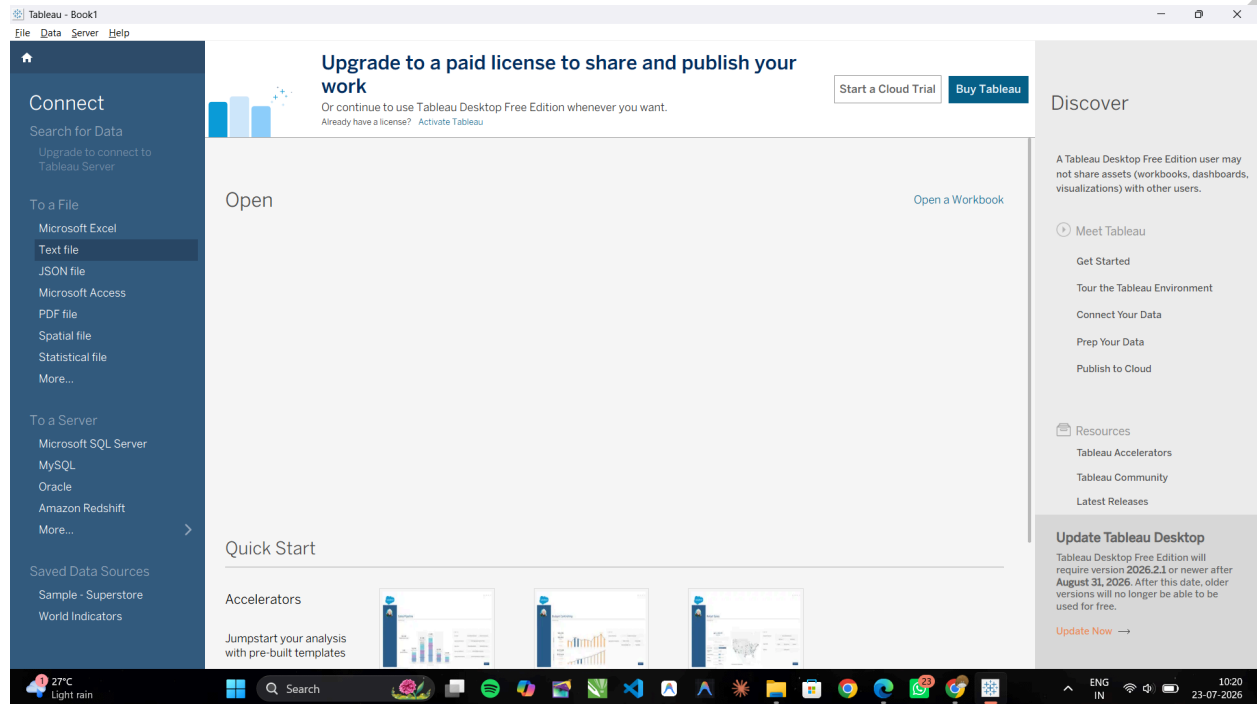
Dataset: ipl.csv

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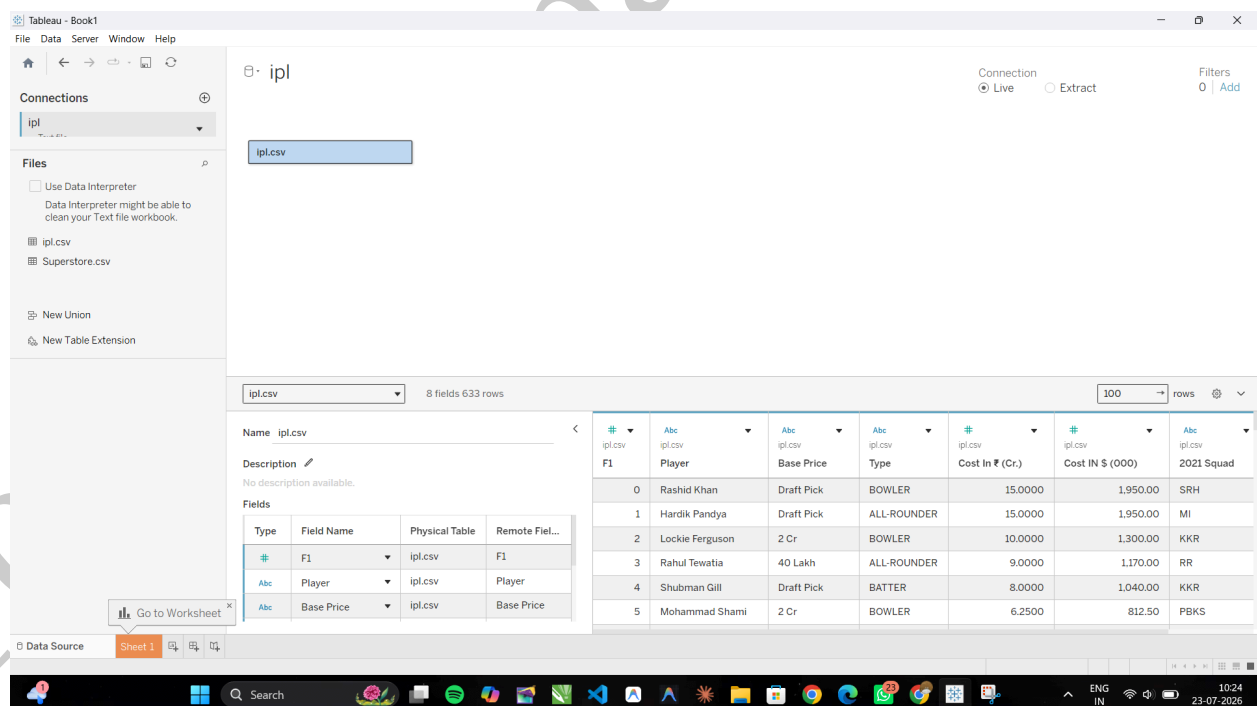
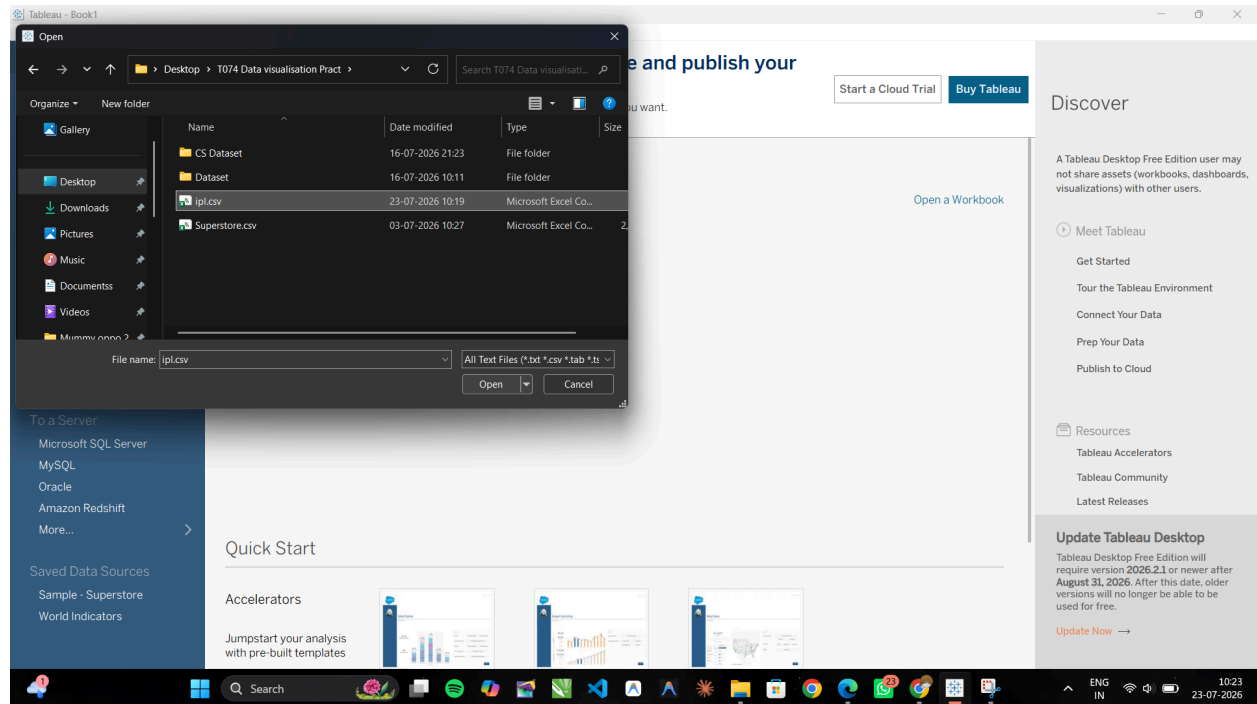
Step A: Creating Stacked and Grouped Bar Charts for Comparison

1. Launch Tableau. Under the Connect panel, click on Text File, select your Csv file, and click Open.



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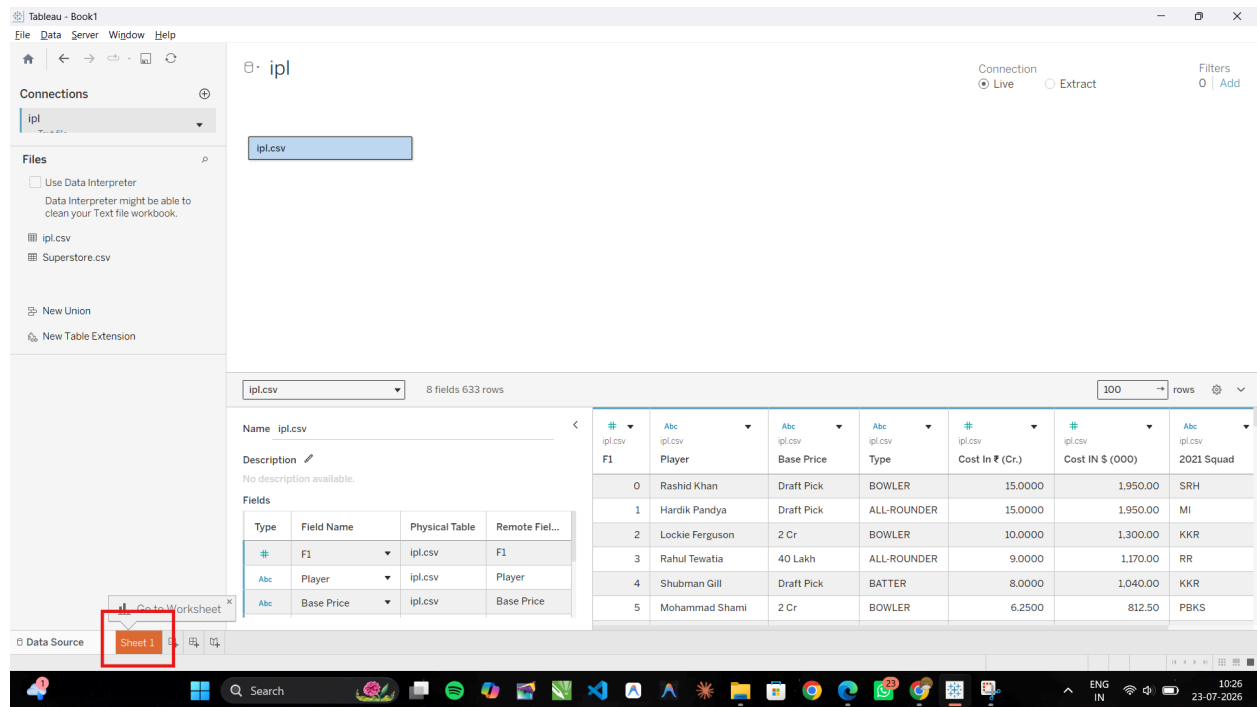
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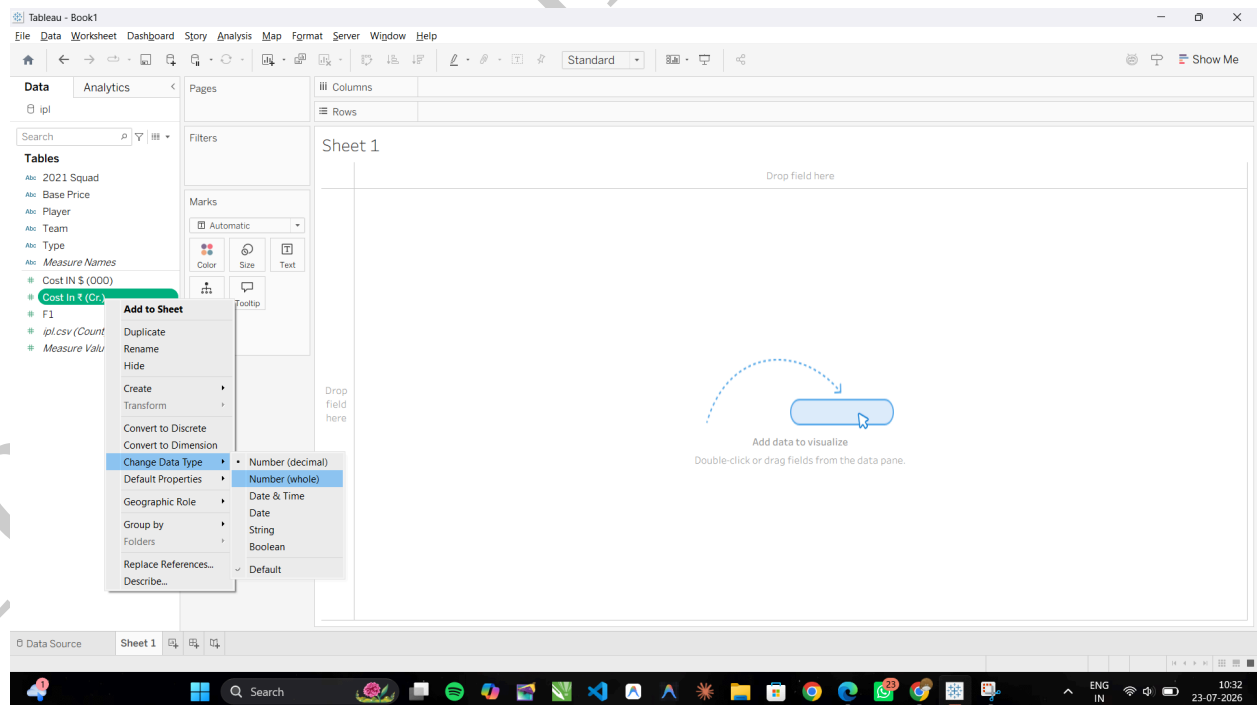
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Click on **Sheet 1**



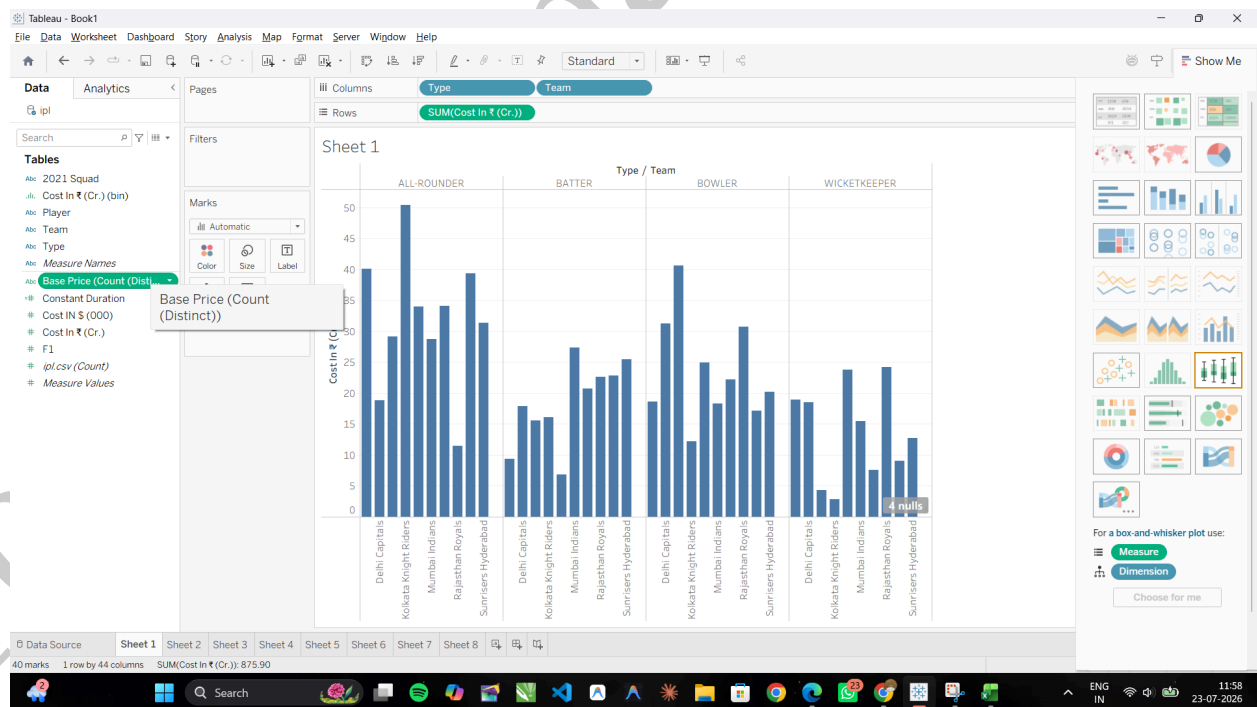
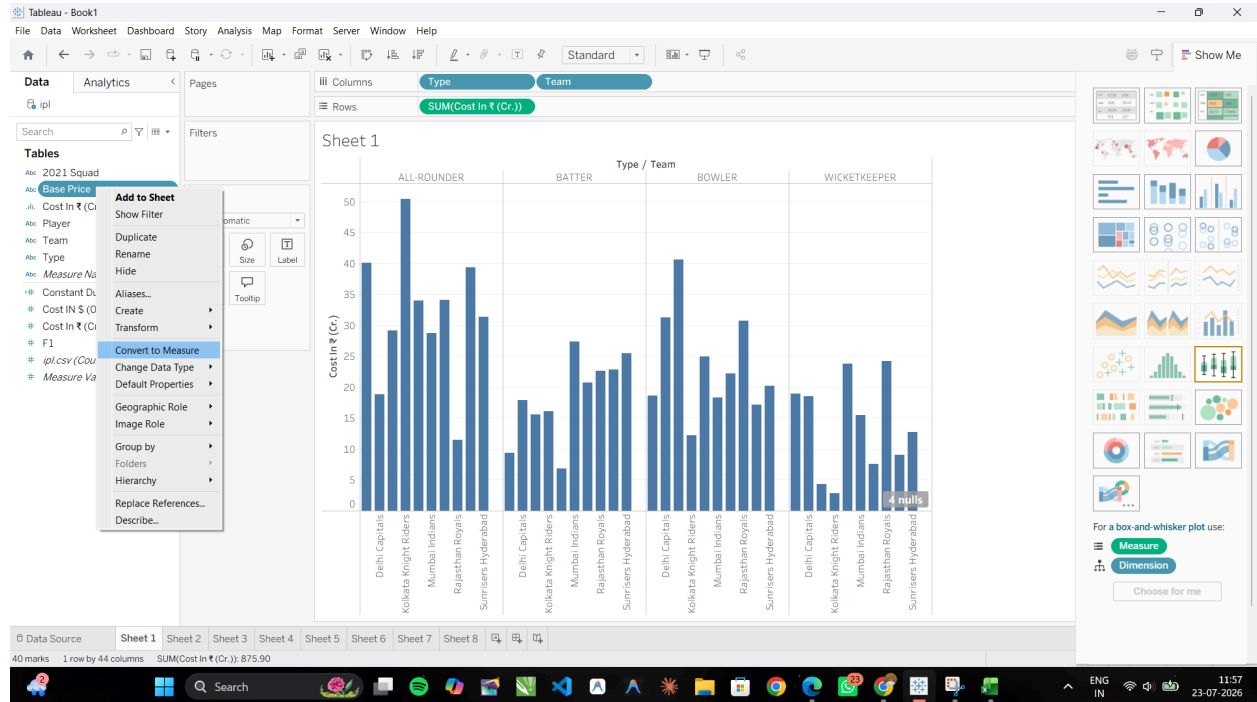
Fix the Data Type: In the Data pane on the left, locate Amount in USD. Click the small Abc text icon next to it and select Number (Whole) from the dropdown menu to convert it into numeric data.



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Convert to Measure: Right-click Base Price in the Data pane and select Convert to Measure. It will turn green and move to the bottom half of the pane.

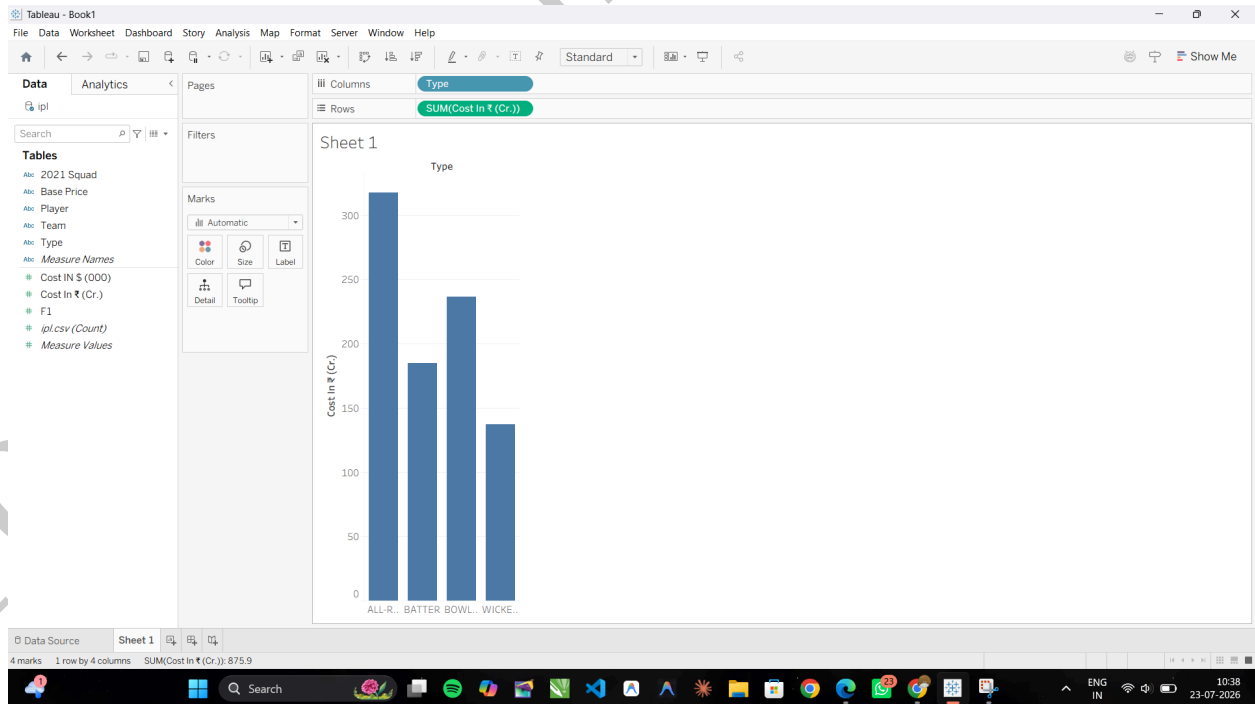
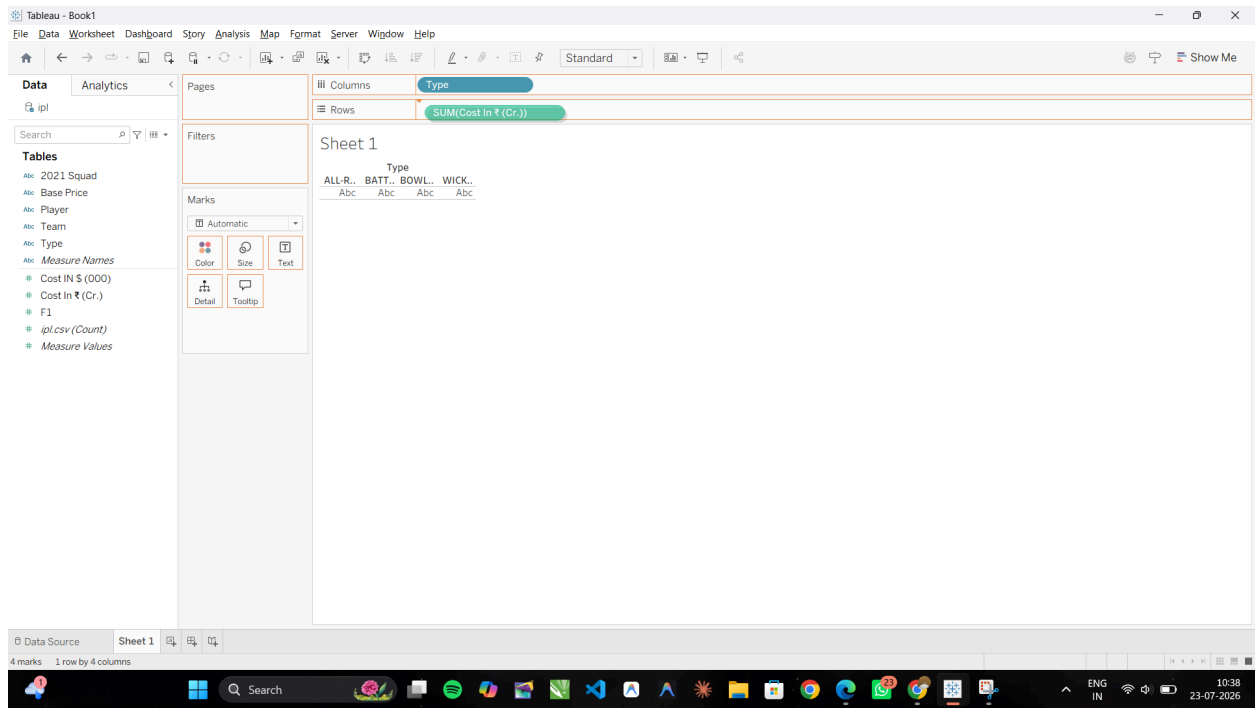


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2. Building a Stacked Bar Chart

From the **Data** pane, drag the **Team** dimension and drop it onto the **Columns** shelf and **Cost In ₹ (Cr.)** Rows shelf.



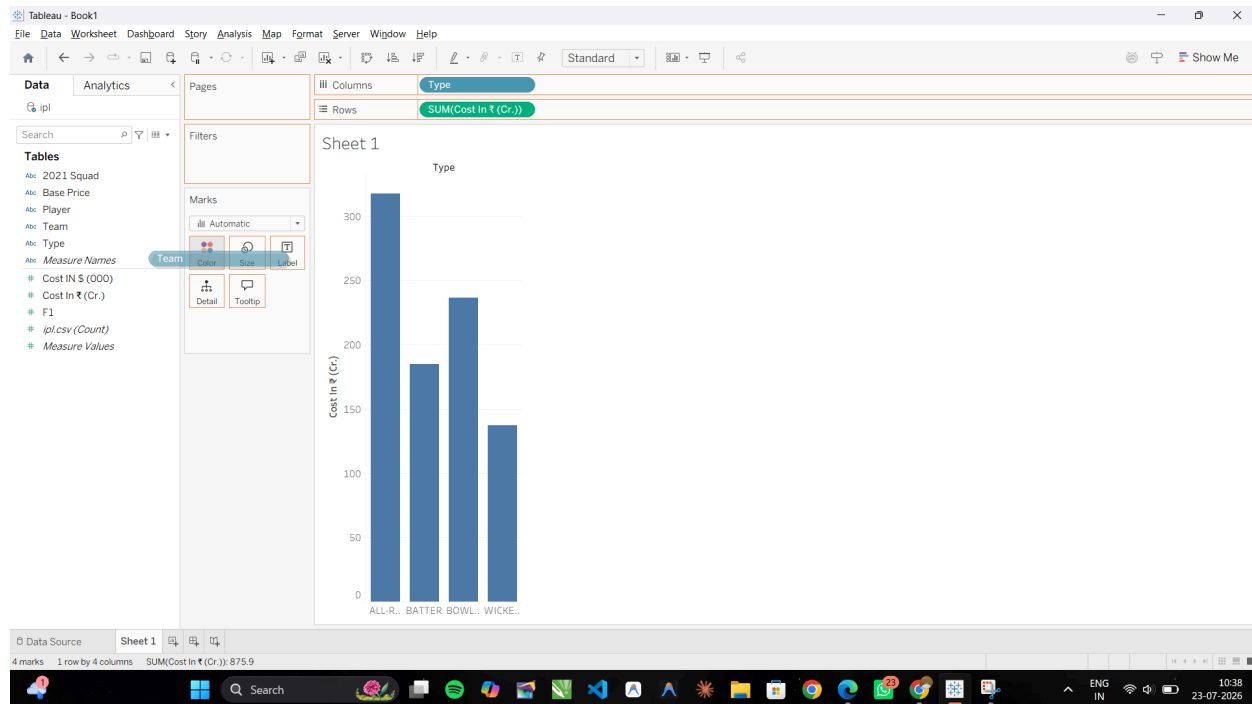
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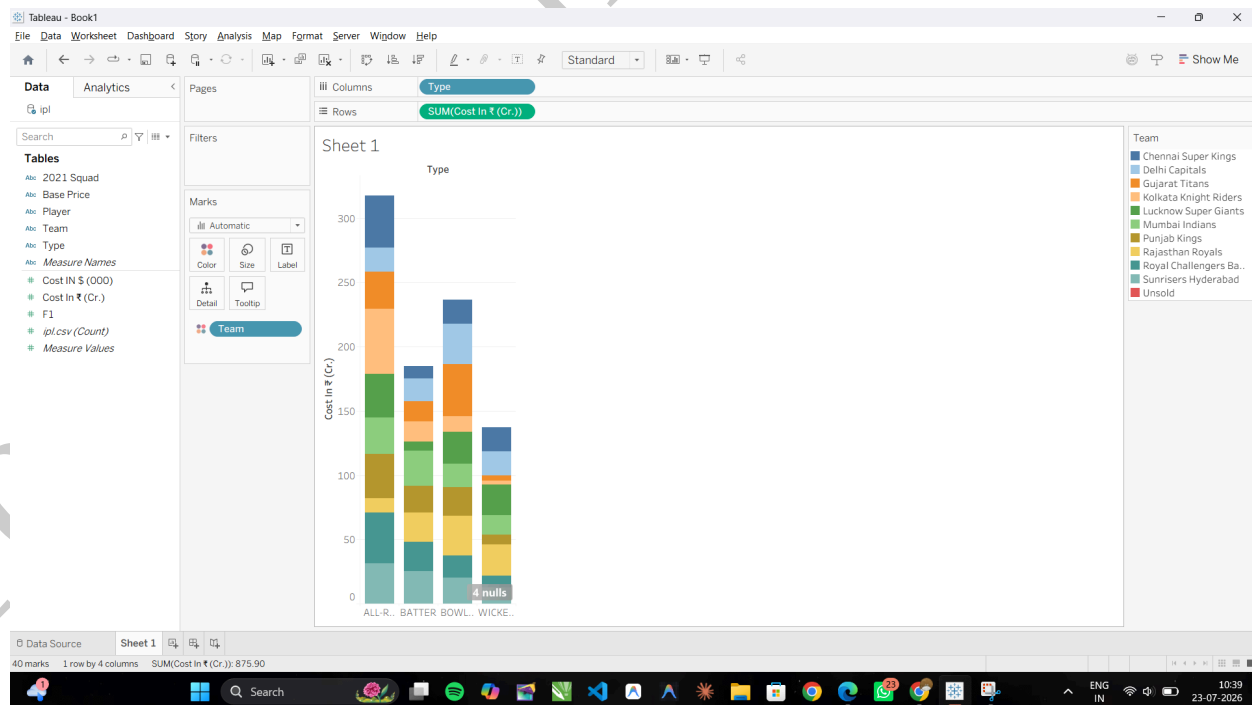
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Drag the **Type** dimension onto the **Color** option in the **Marks** card. Tableau automatically creates a **Stacked Bar Chart** showing the total player cost for each team, divided according to player type.



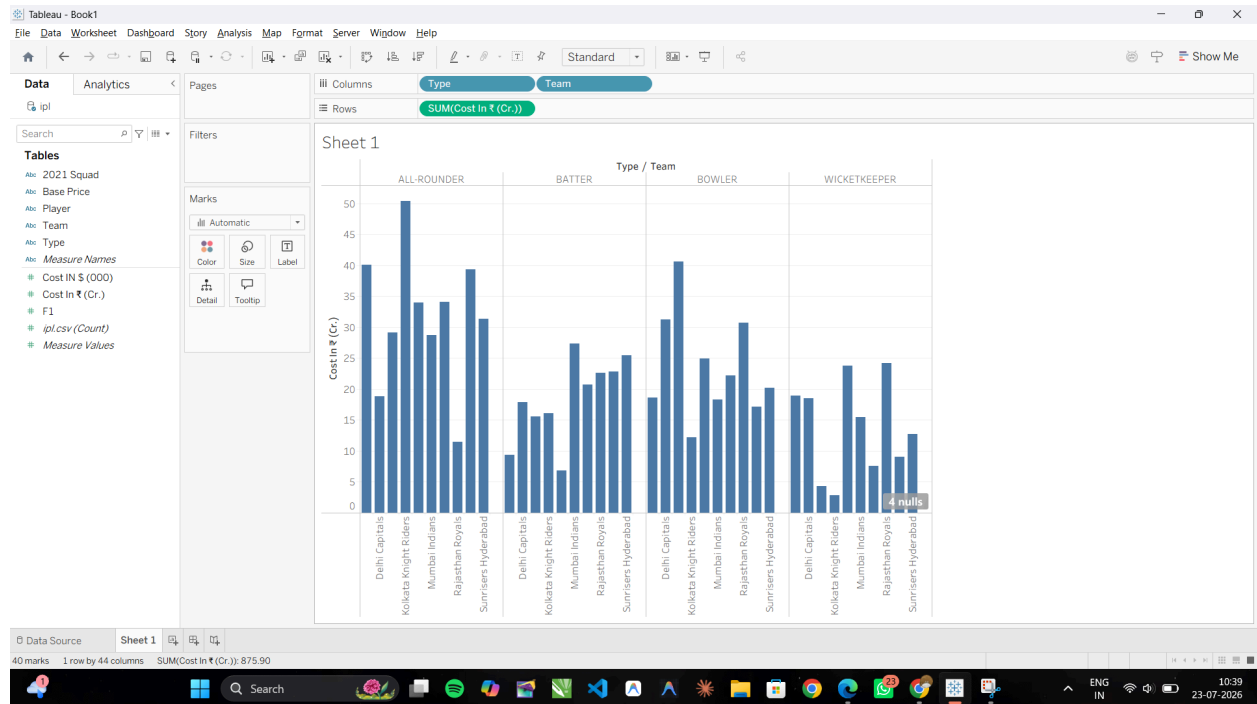
To create a **Grouped Bar Chart**, go to **Analysis** → **Stack Marks** → **Off**.



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If required, sort the teams by clicking the **Sort Descending** icon on the toolbar.



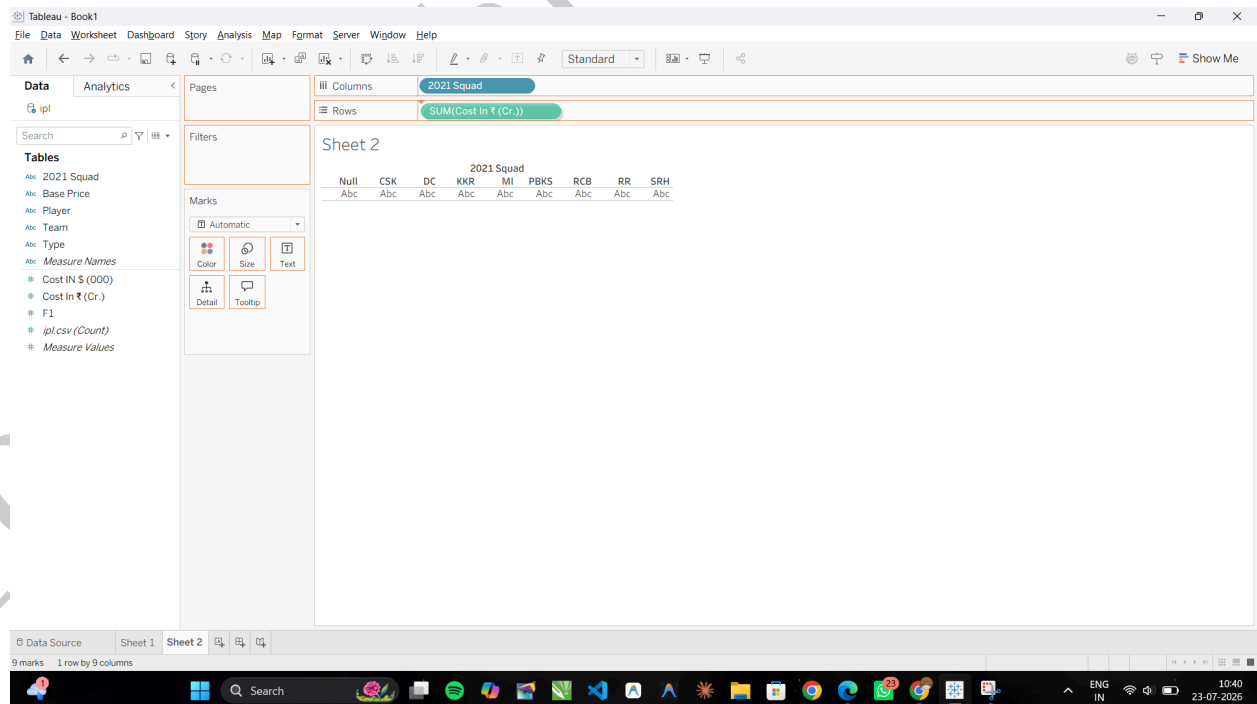
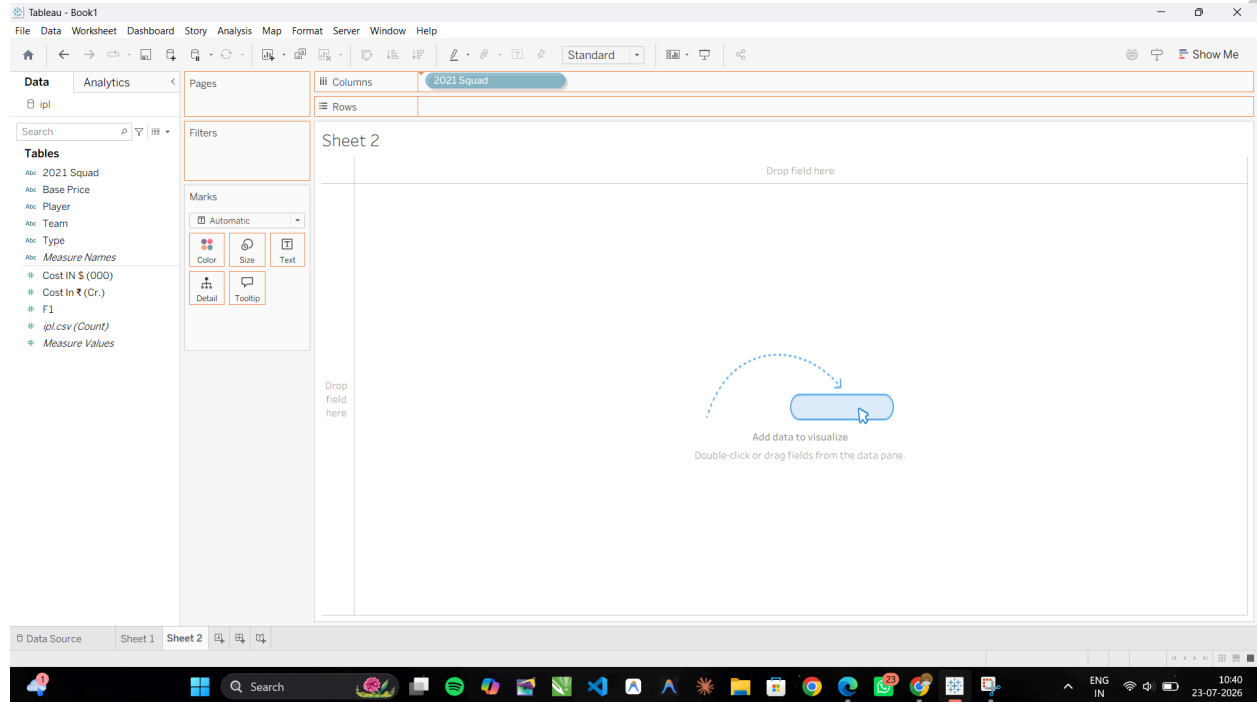
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Step B: Building Treemaps and Pie Charts for Part-to-Whole Analysis

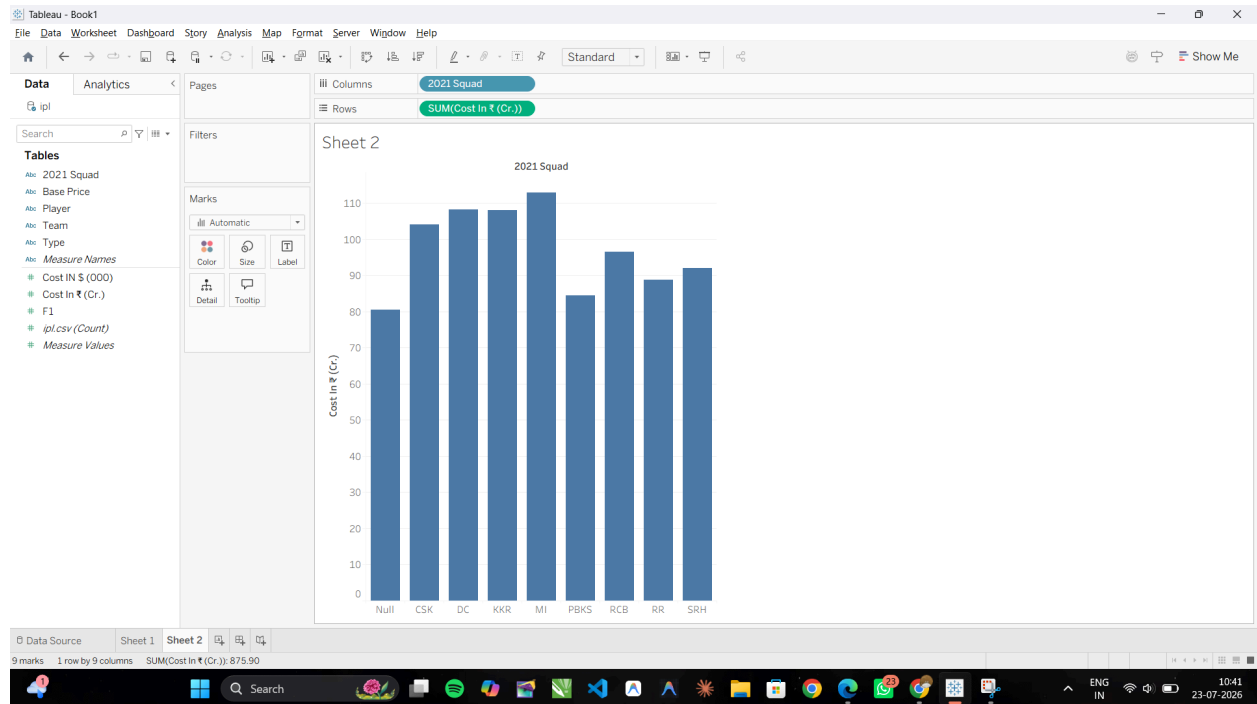
1. Building a Treemap

Click the New Worksheet icon at the bottom of the screen to open a clean workspace (Sheet 2) and Drag **2021 Squad** into **Columns** and **Cost In ₹ (Cr.)** in **Rows**

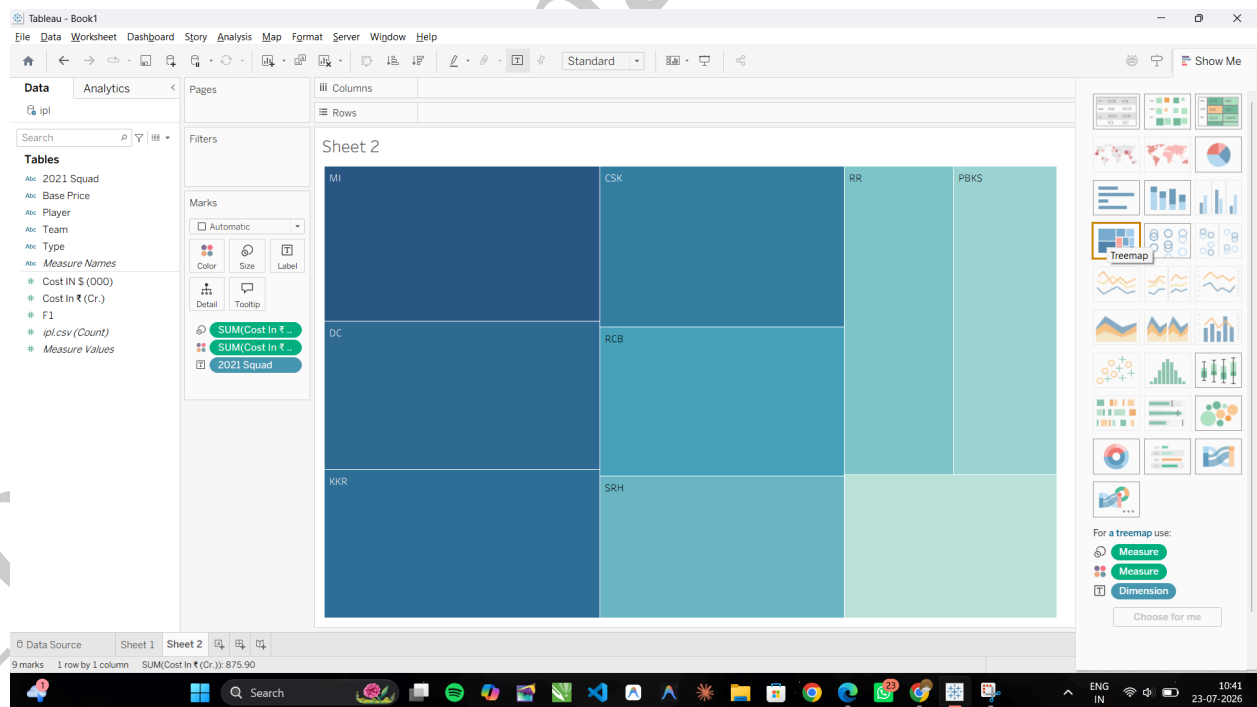


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Click the **Show Me** button and select the **Treemap** chart.

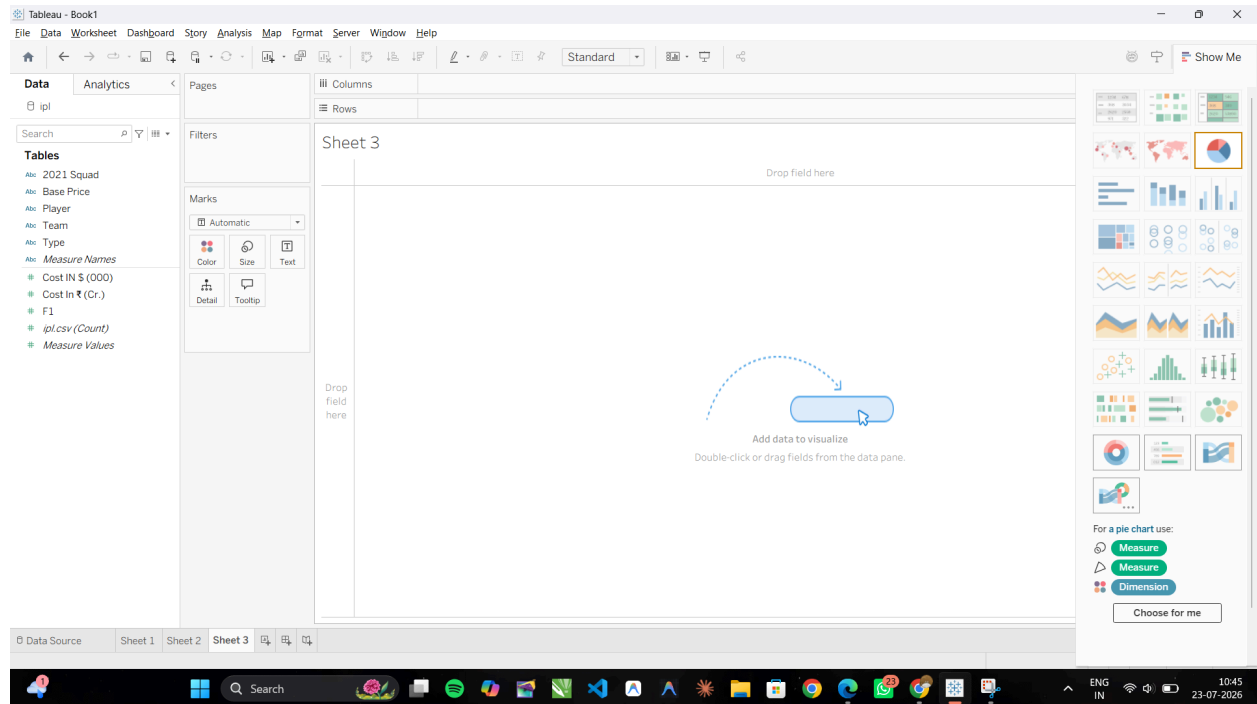


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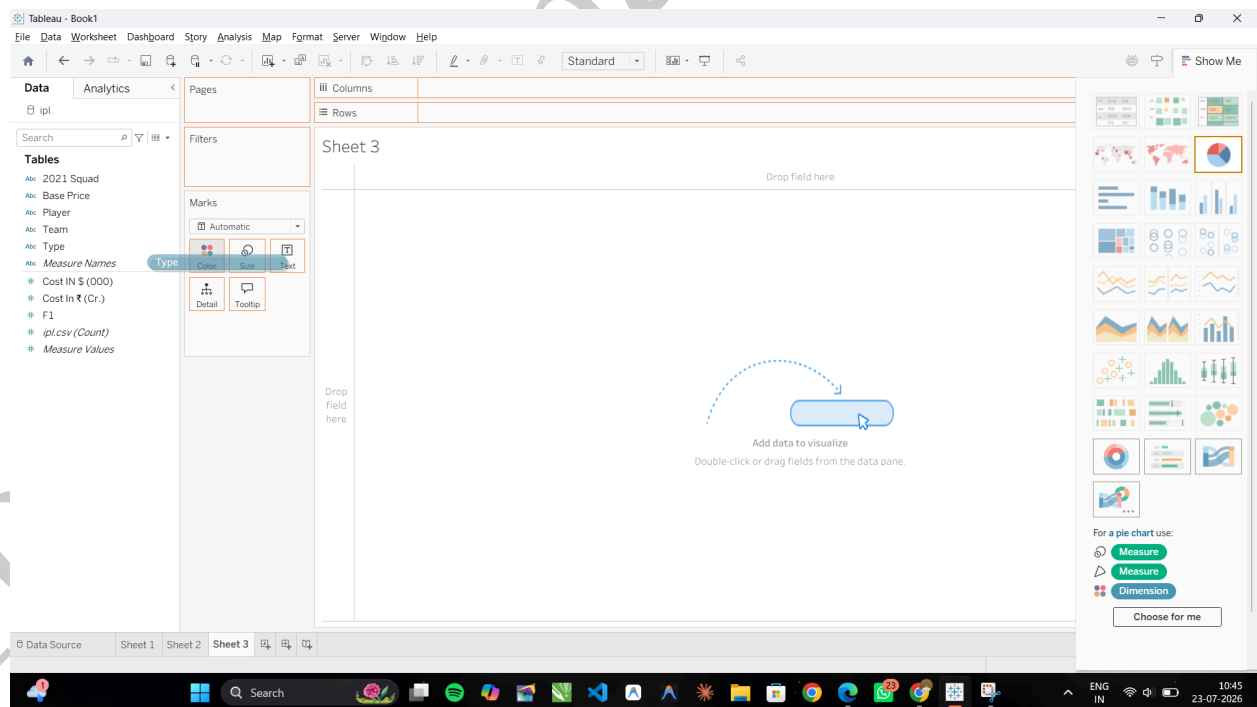
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2. Building a Pie Chart

Click the **New Worksheet** icon to open another blank layout (Sheet 3).

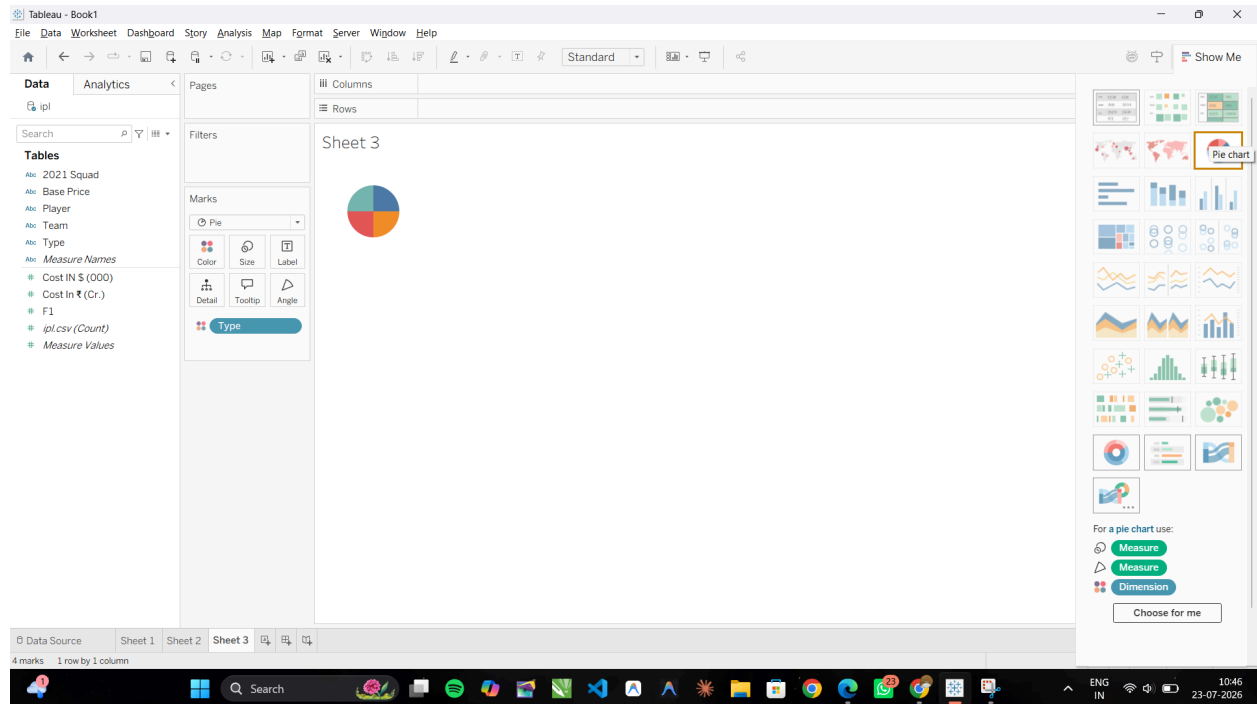


Drag **Type** to the **Color** option in the **Marks** card.

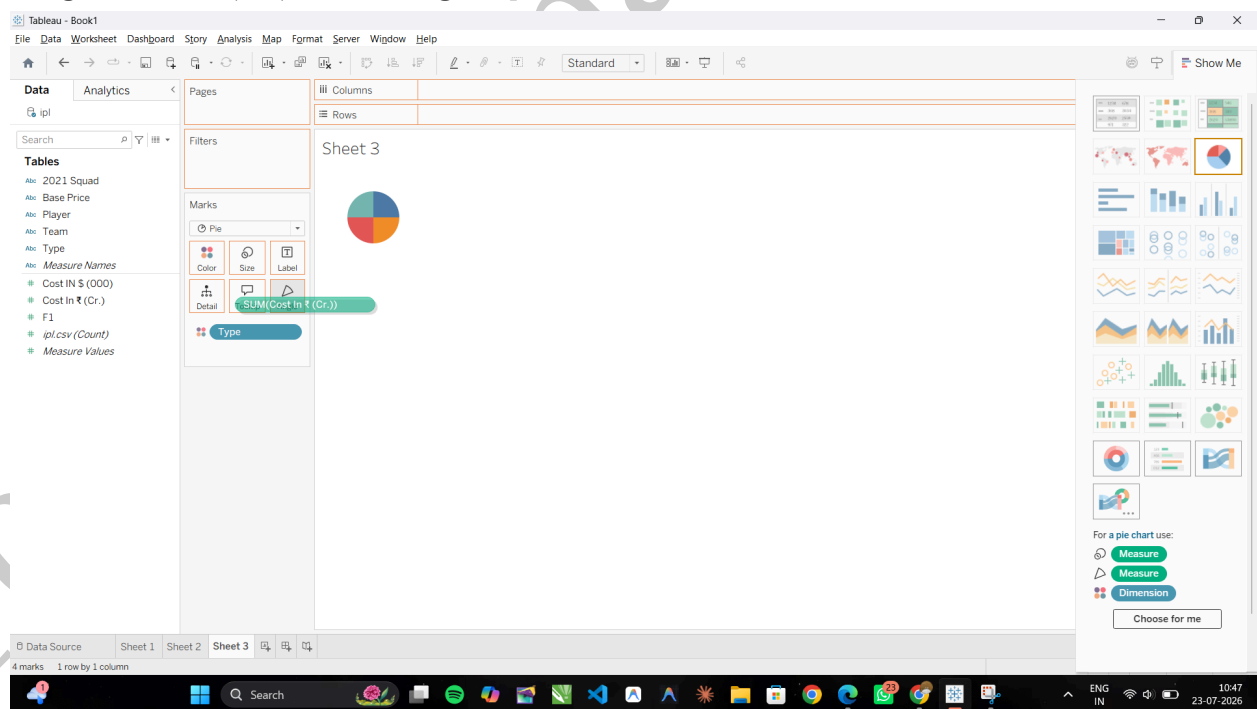


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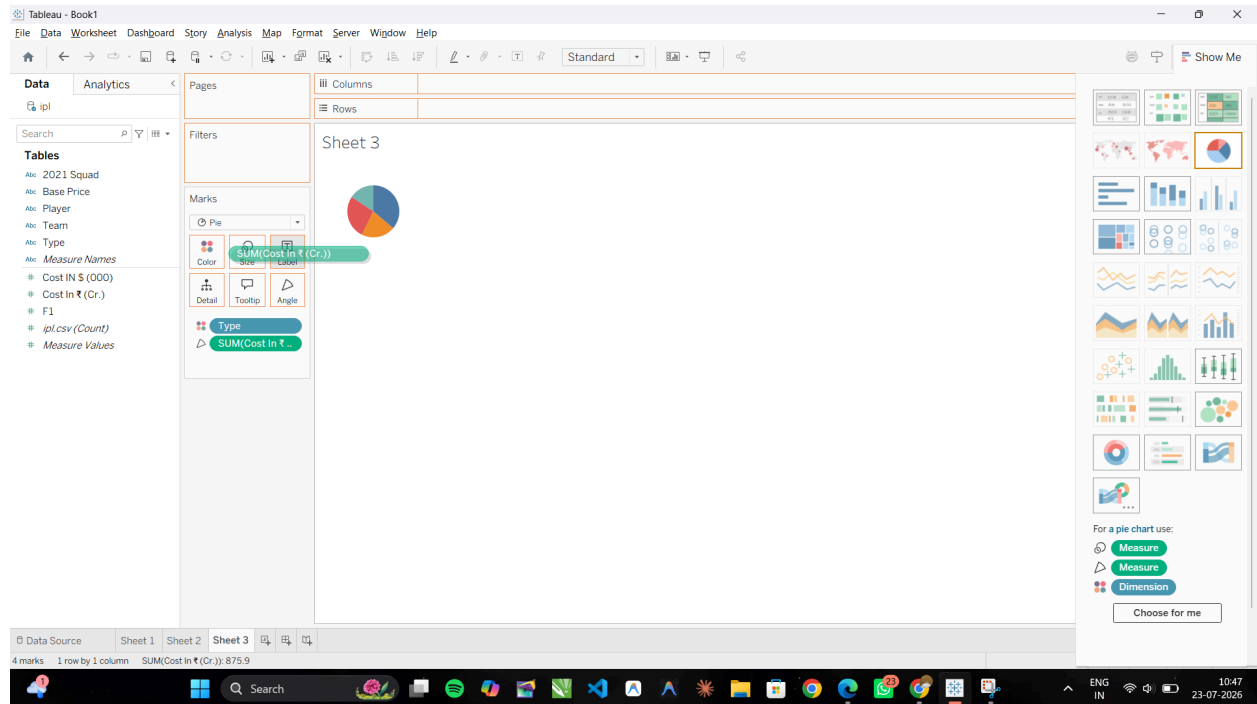
Drag **Cost In ₹ (Cr.)** to the **Angle** option in the **Marks** card.



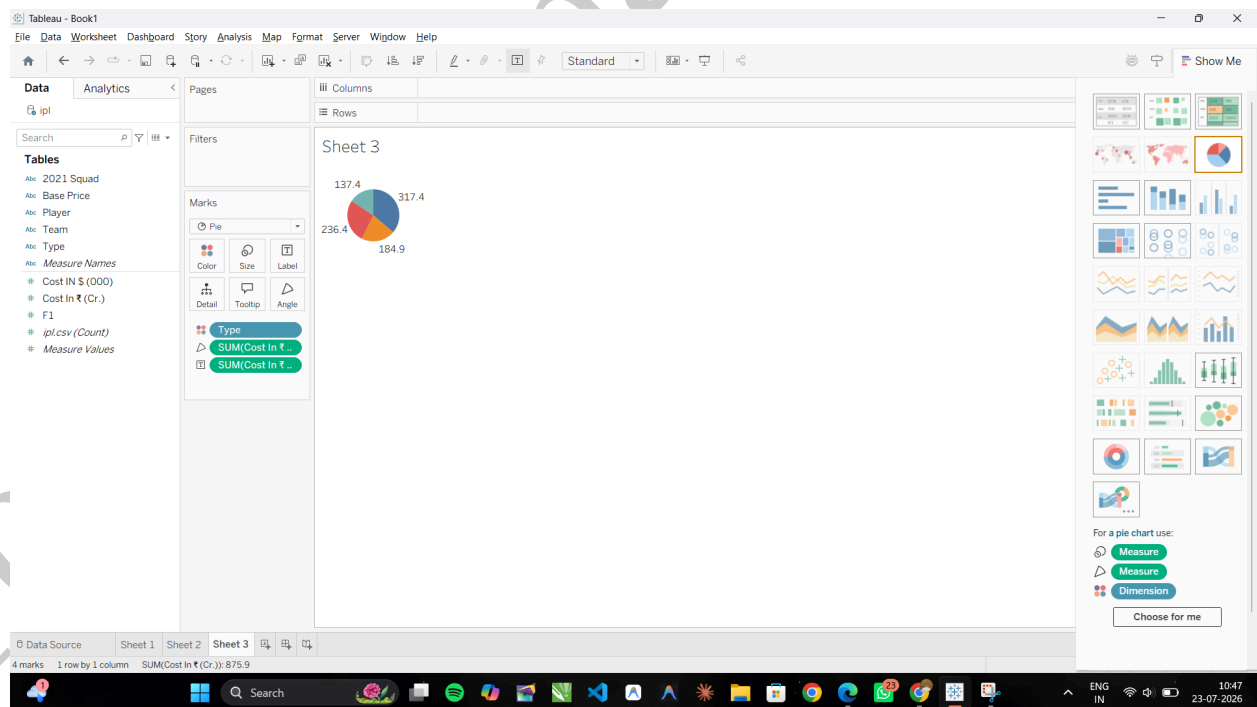
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Drag **Type** to the **Label** option in the **Marks** card.



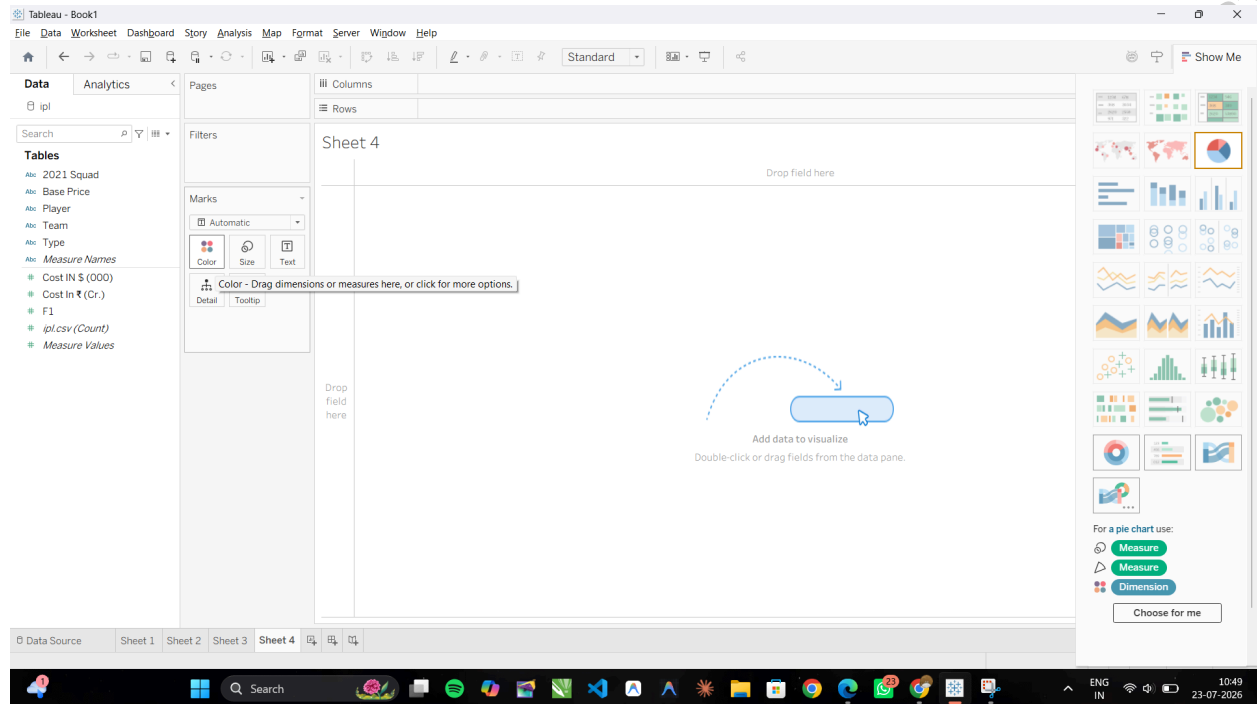
Click **Show Me** and select the **Pie Chart**.



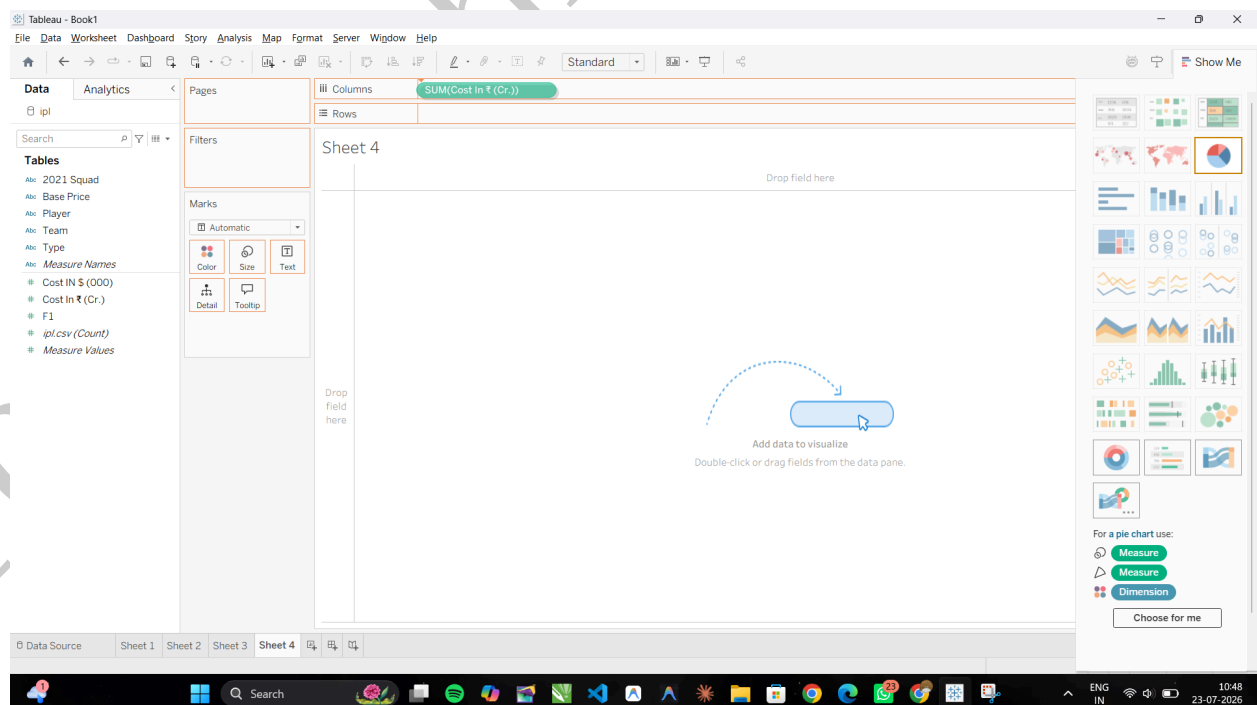
Step C: Designing Scatter Plots to Identify Relationships

1. Cleaning the Data for Scatter Plots

Click the New Worksheet icon at the bottom of your window to open a fresh canvas (Sheet 4).



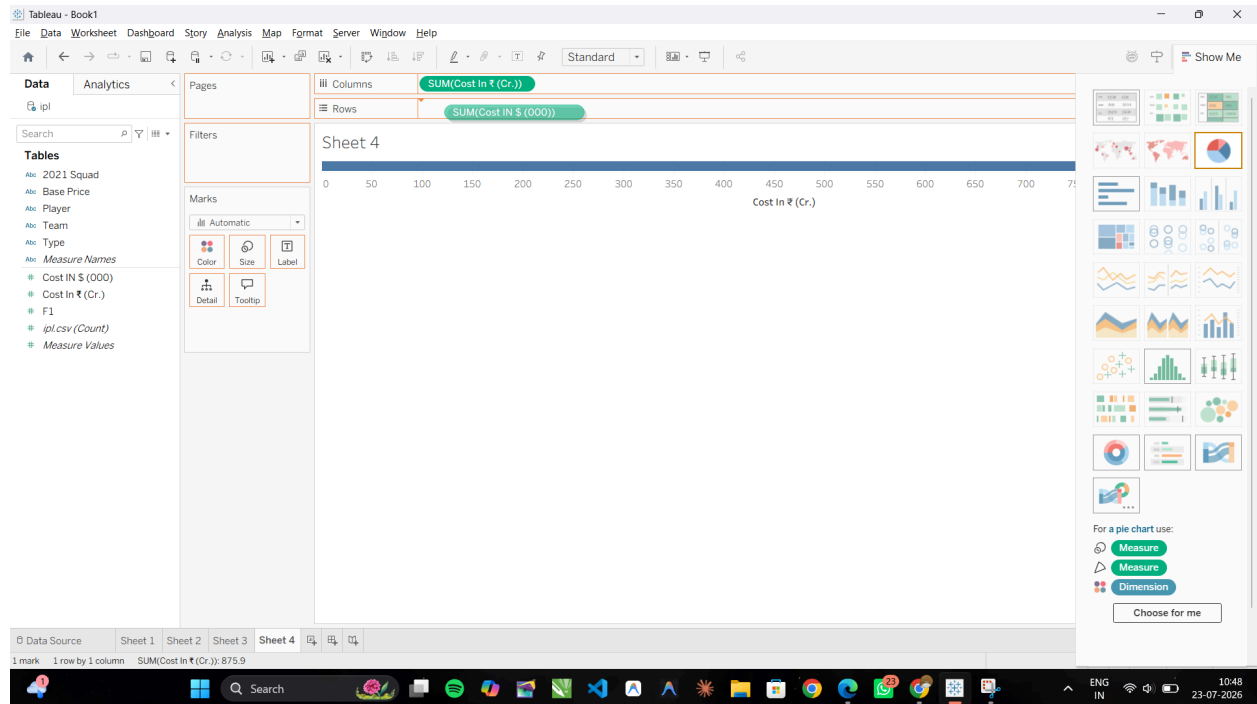
Drag Cost In ₹ (Cr.) to the Columns shelf.



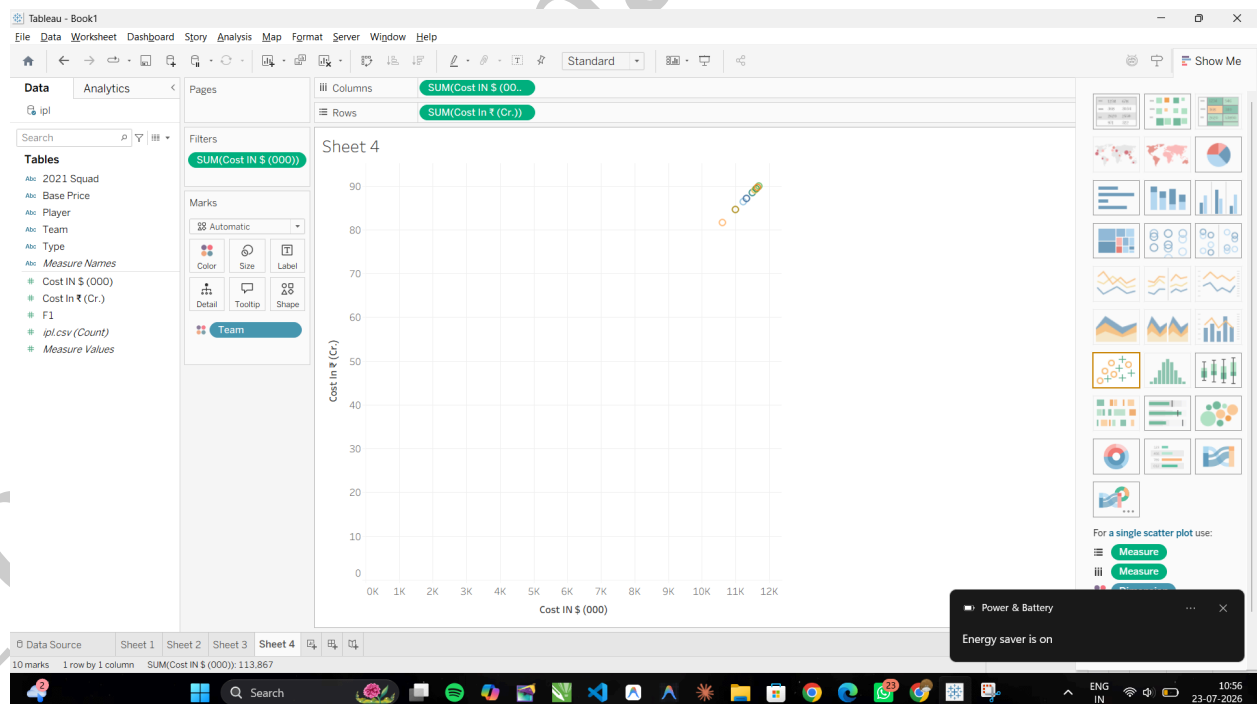
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Drag Cost In \$ (000) to the Rows shelf.



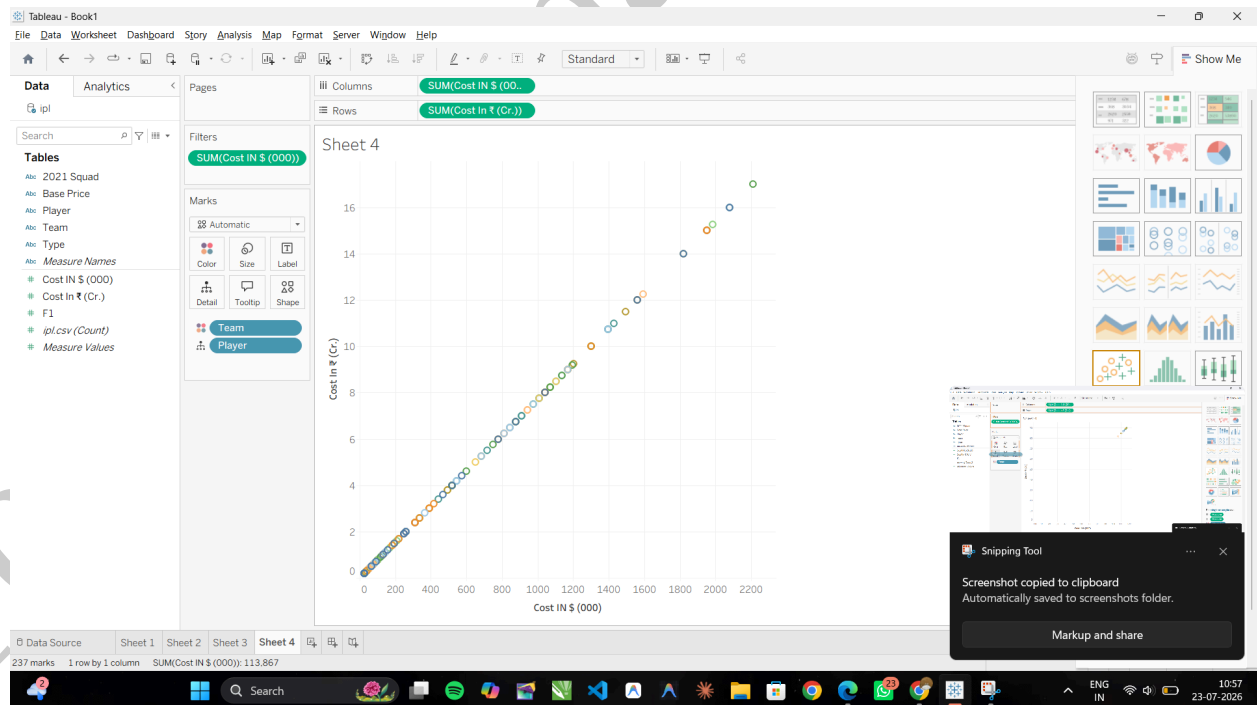
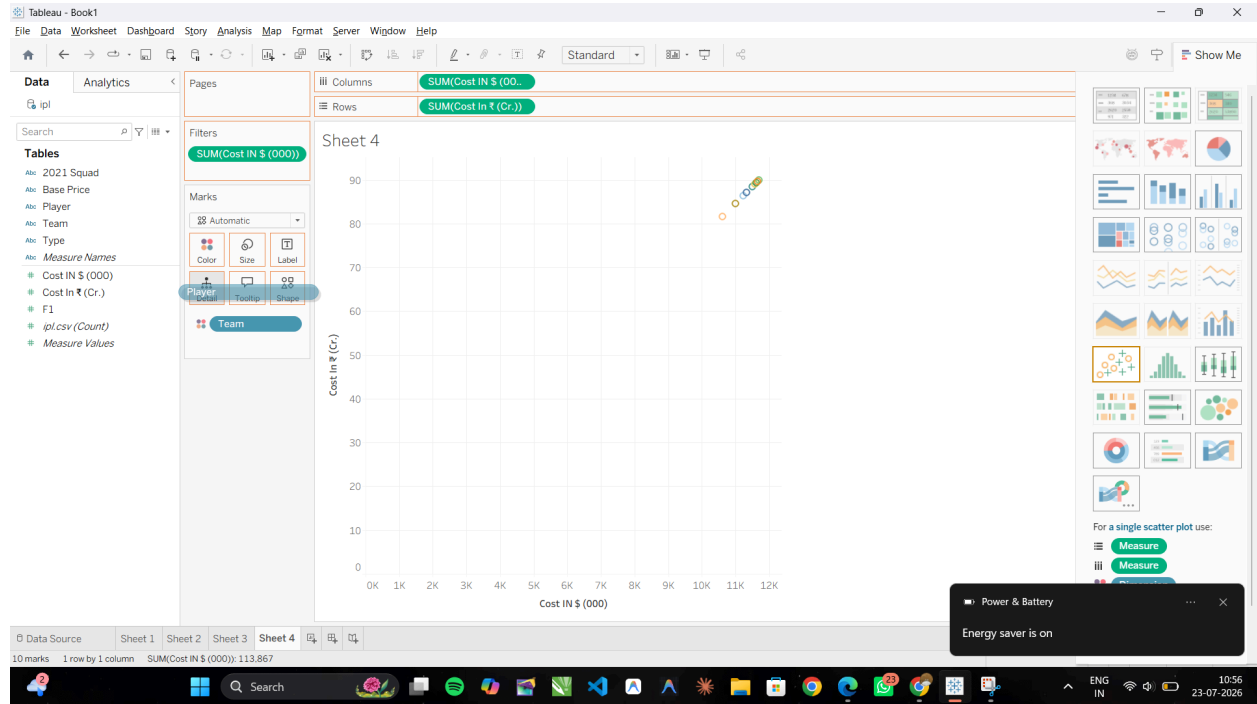
Drag **Team** to the **Color** option in the **Marks** card to distinguish player types.



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Drag **Player** to the **Detail** option in the **Marks** card so that each player appears as an individual point.

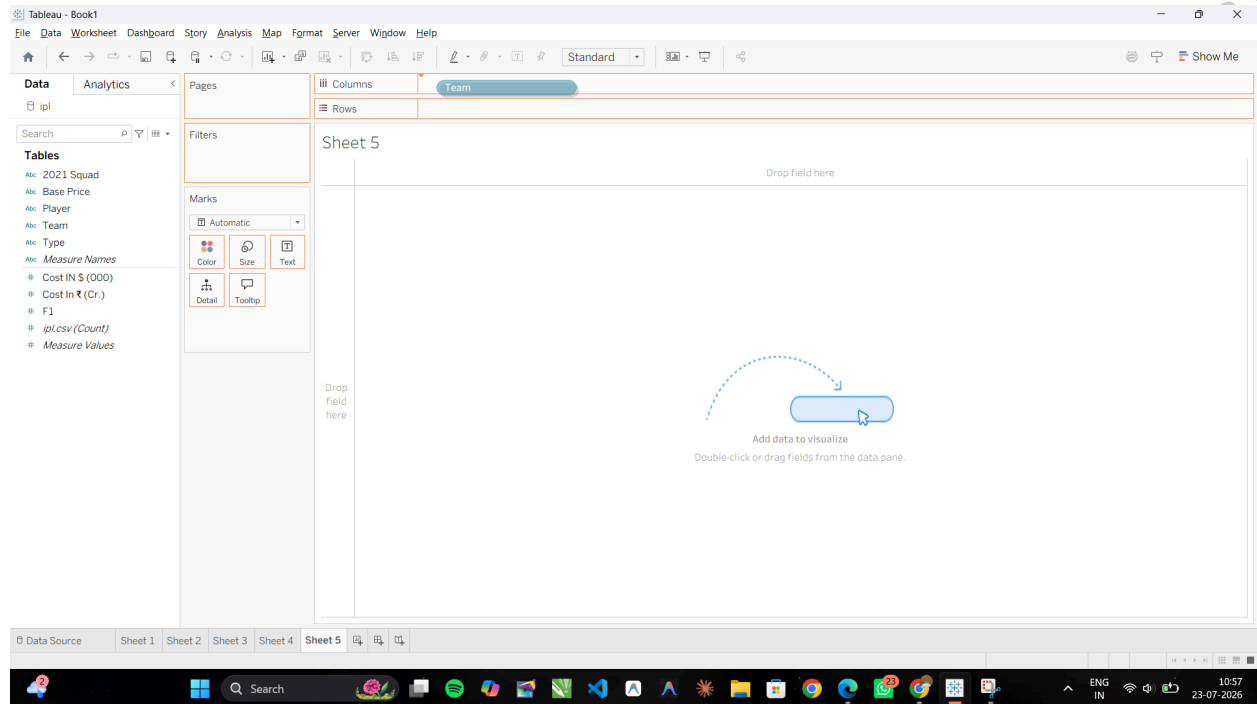


Step D: Creating Dual-Axis and Combination Charts

1. Preparing the Fields

Click on the New Worksheet icon to open a new worksheet (Sheet 5).

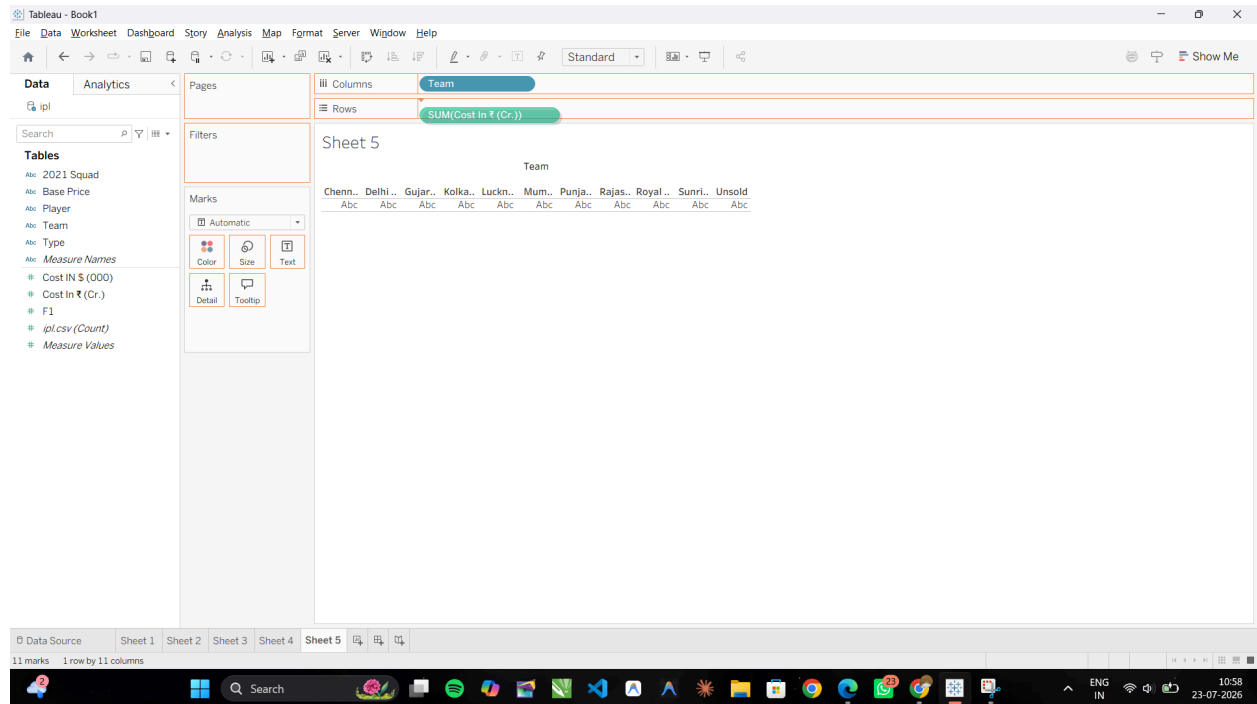
Then, drag Player to the Columns shelf.



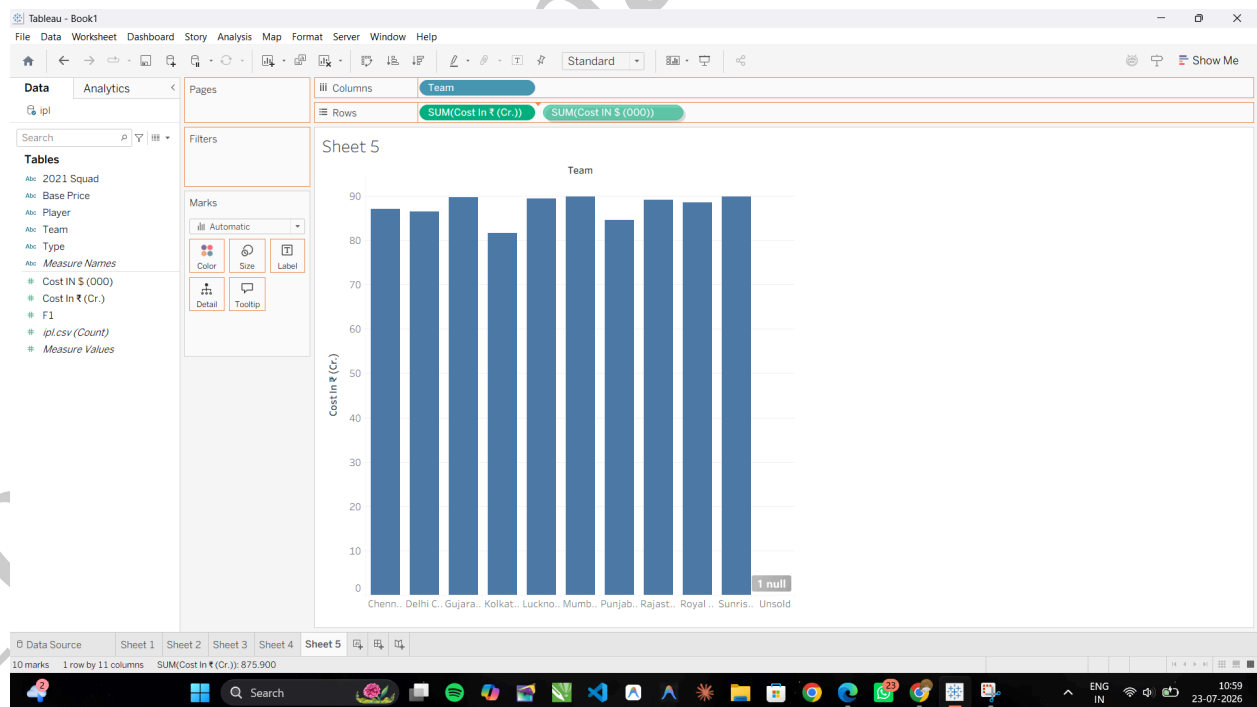
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Drag Cost In ₹ (Cr.) to the Rows shelf.



Drag Cost In \$ (000) and drop it next to Cost In ₹ (Cr.) on the Rows shelf.

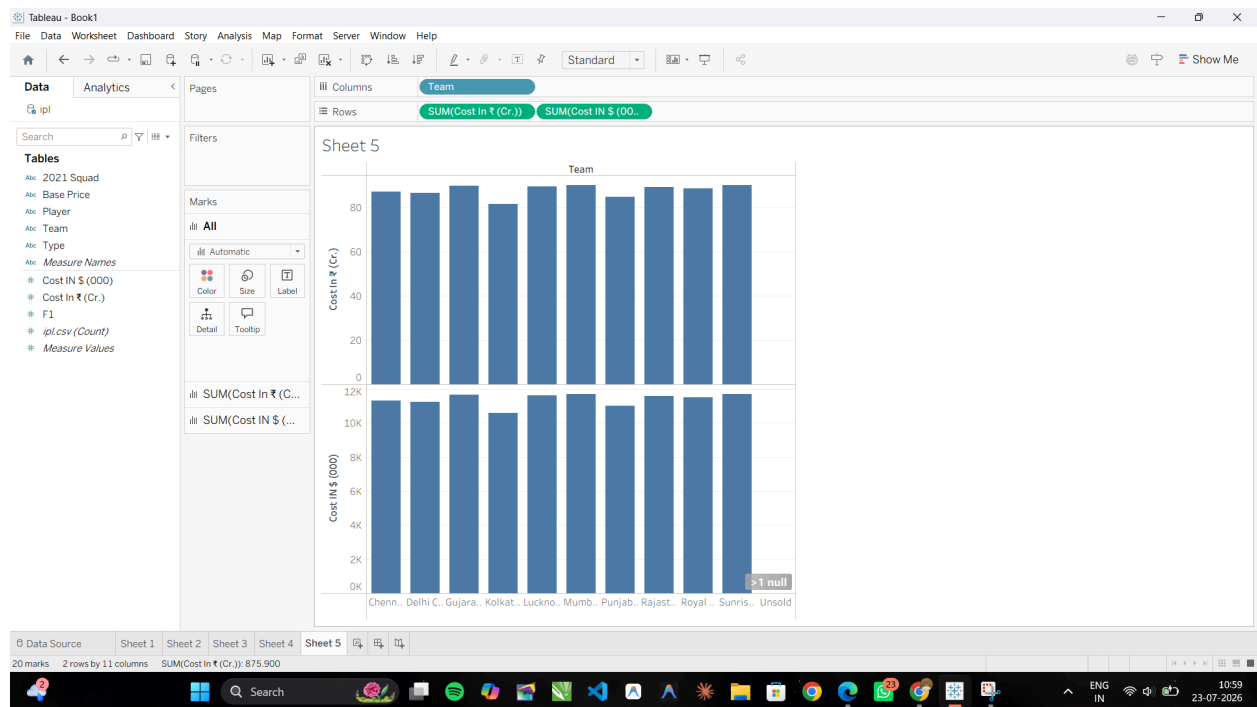


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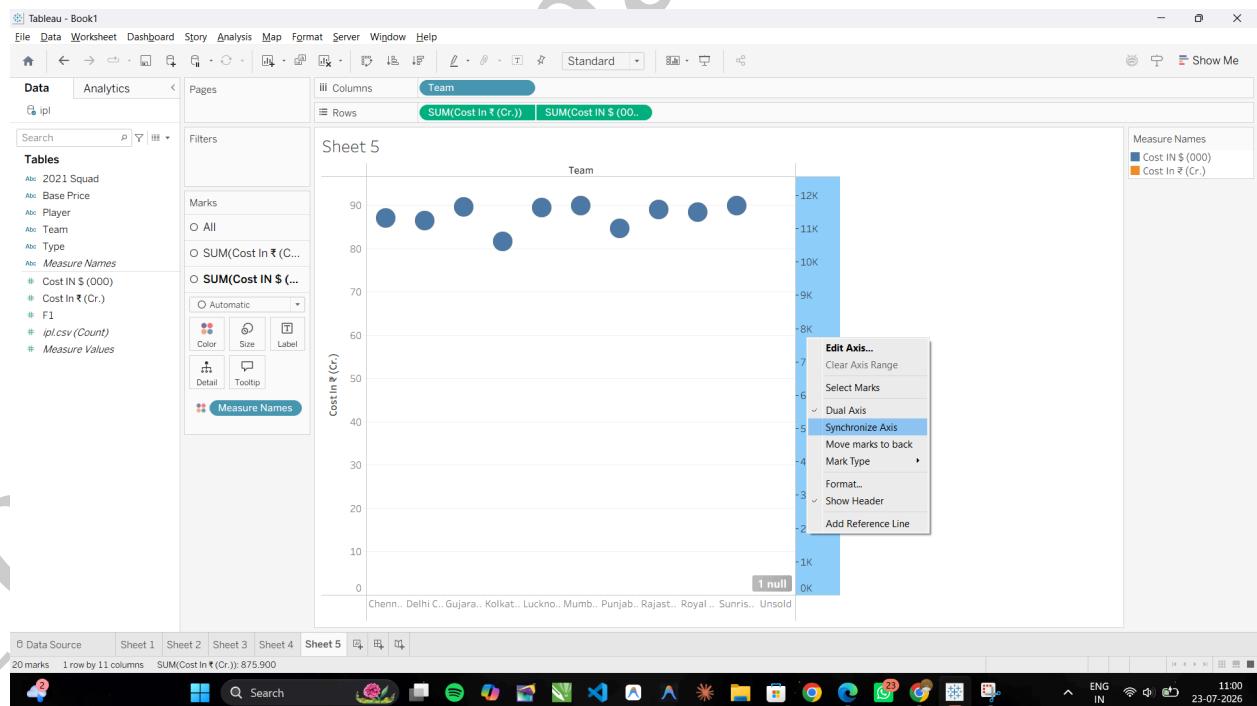
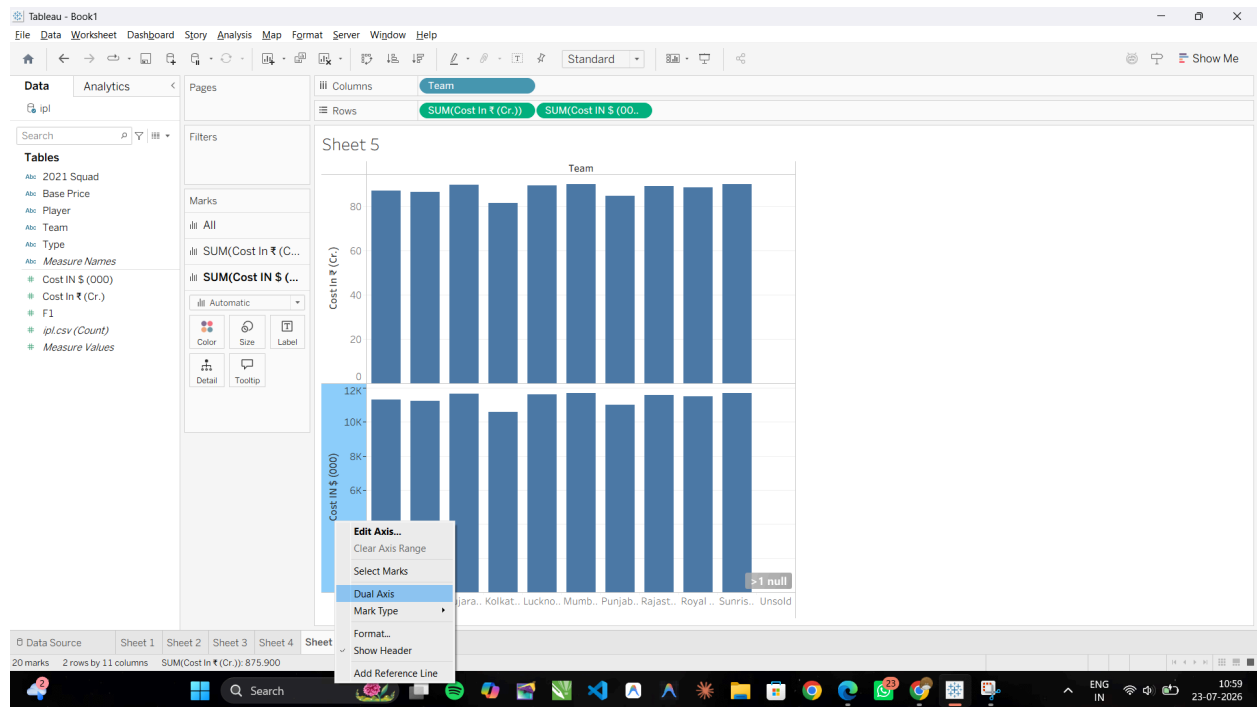
Right-click the second axis (Cost In \$ (000)) and select Dual Axis.



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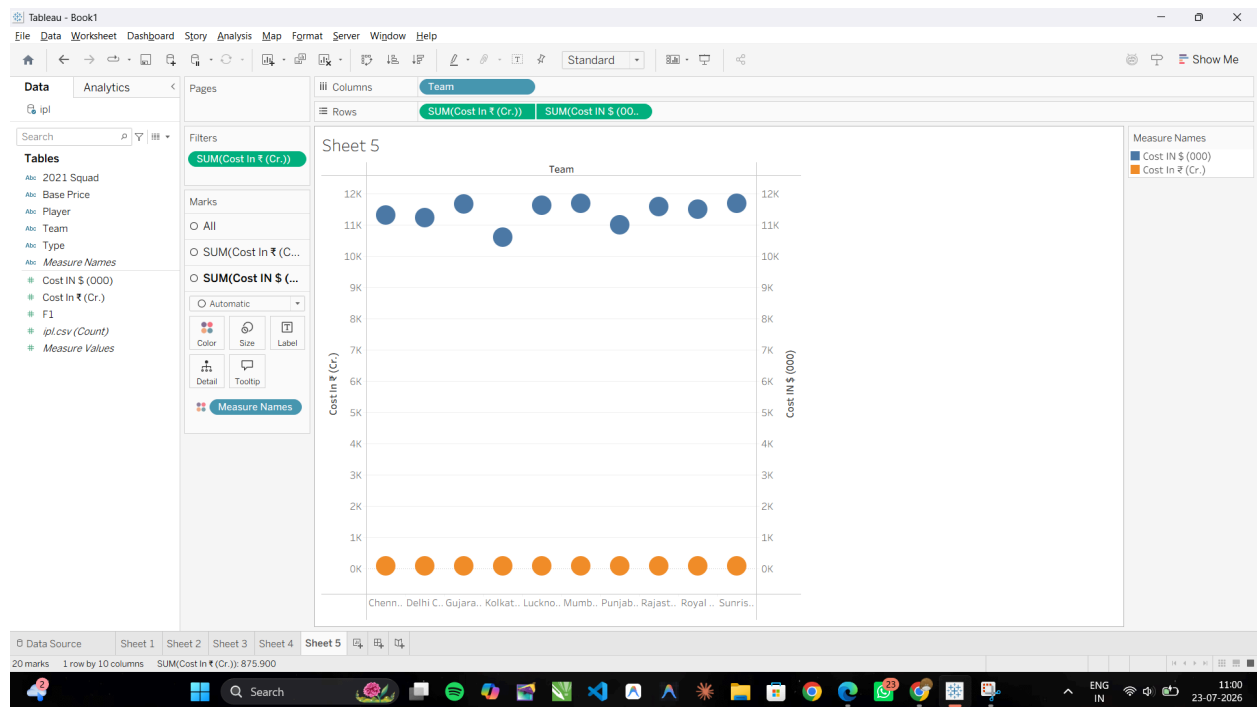
Right-click either axis and select Synchronize Axis to align both measures on the same scale.



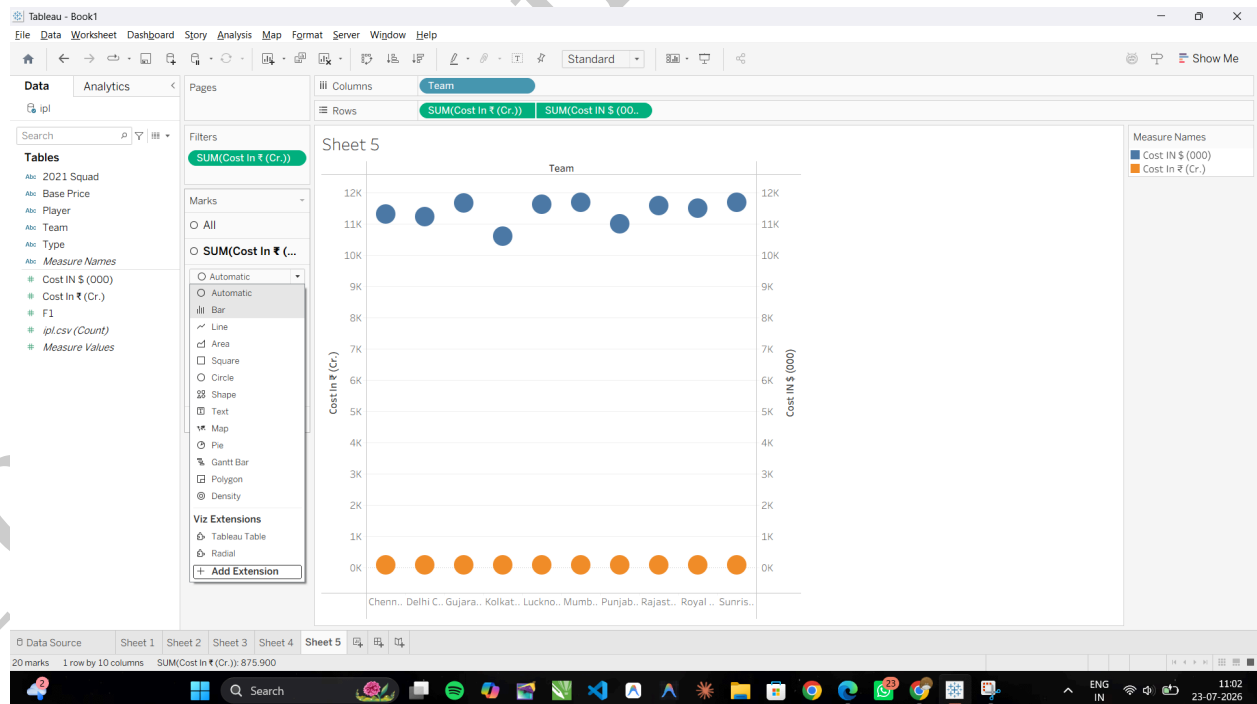
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Drag Measure Names to the Color option in the Marks card if you want to distinguish player types.



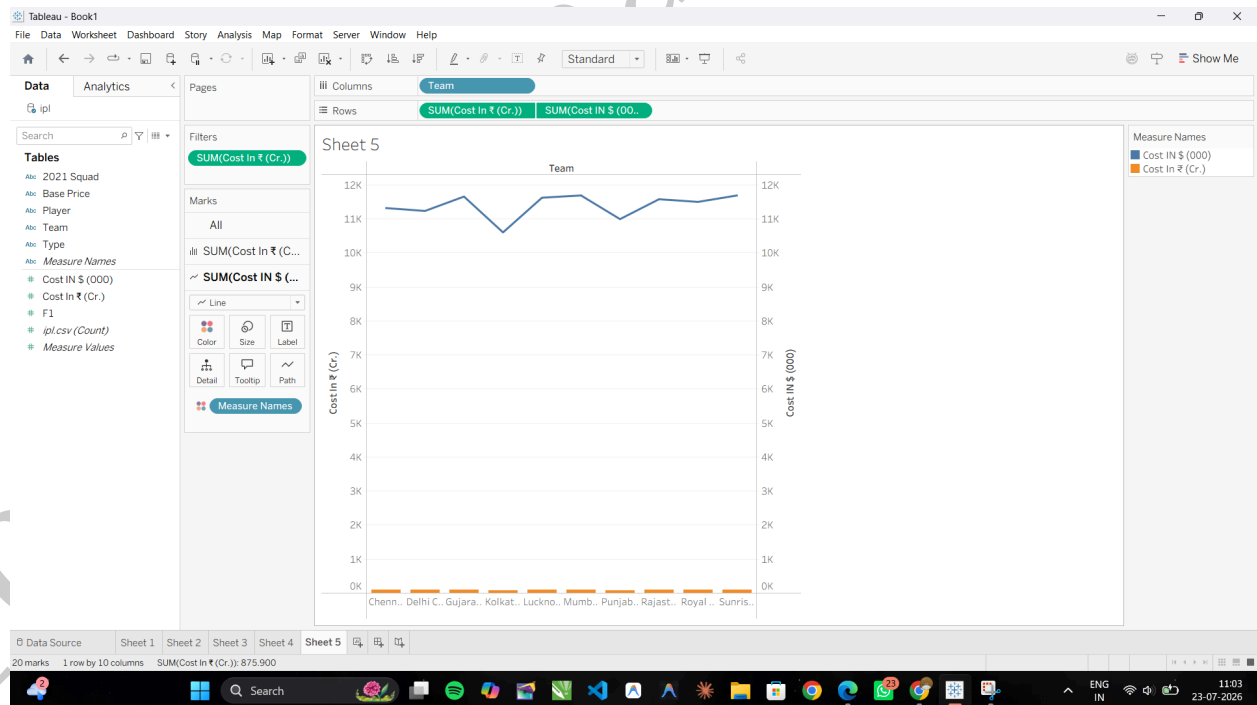
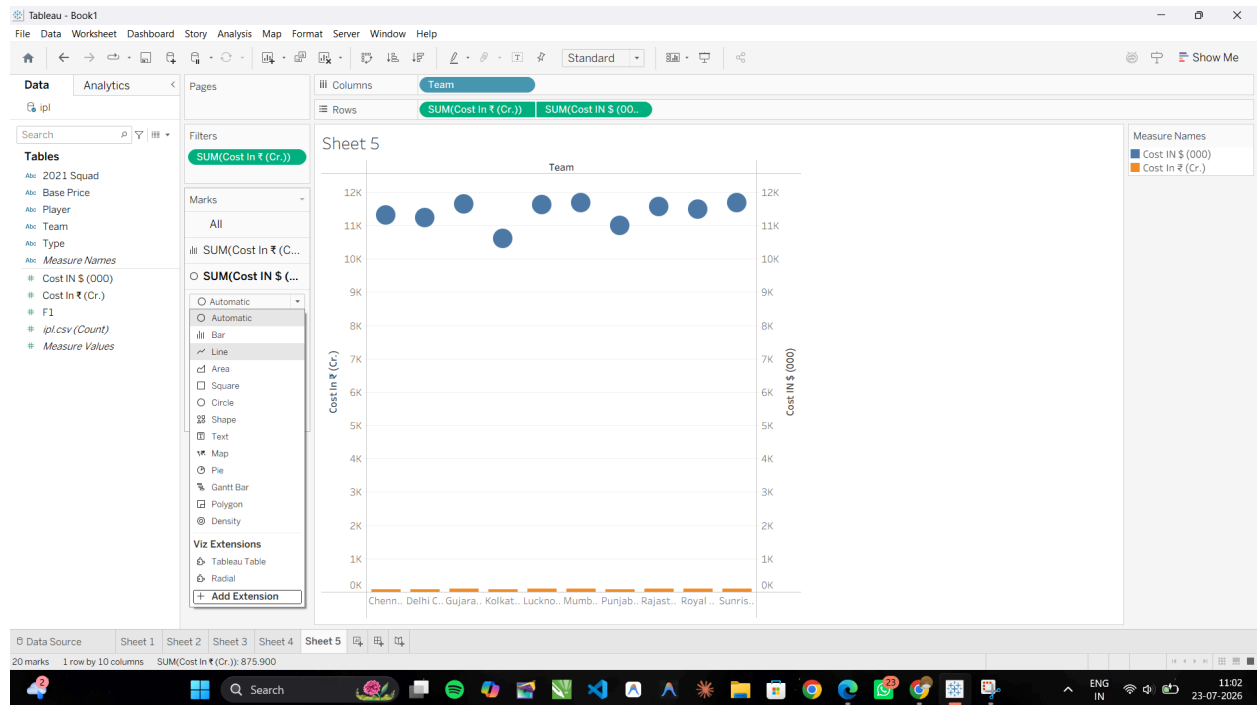
Click the Marks card for Cost In ₹ (Cr.) and change the mark type to Bar.



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Click the Marks card for Cost In \$ (000) and change the mark type to Line.



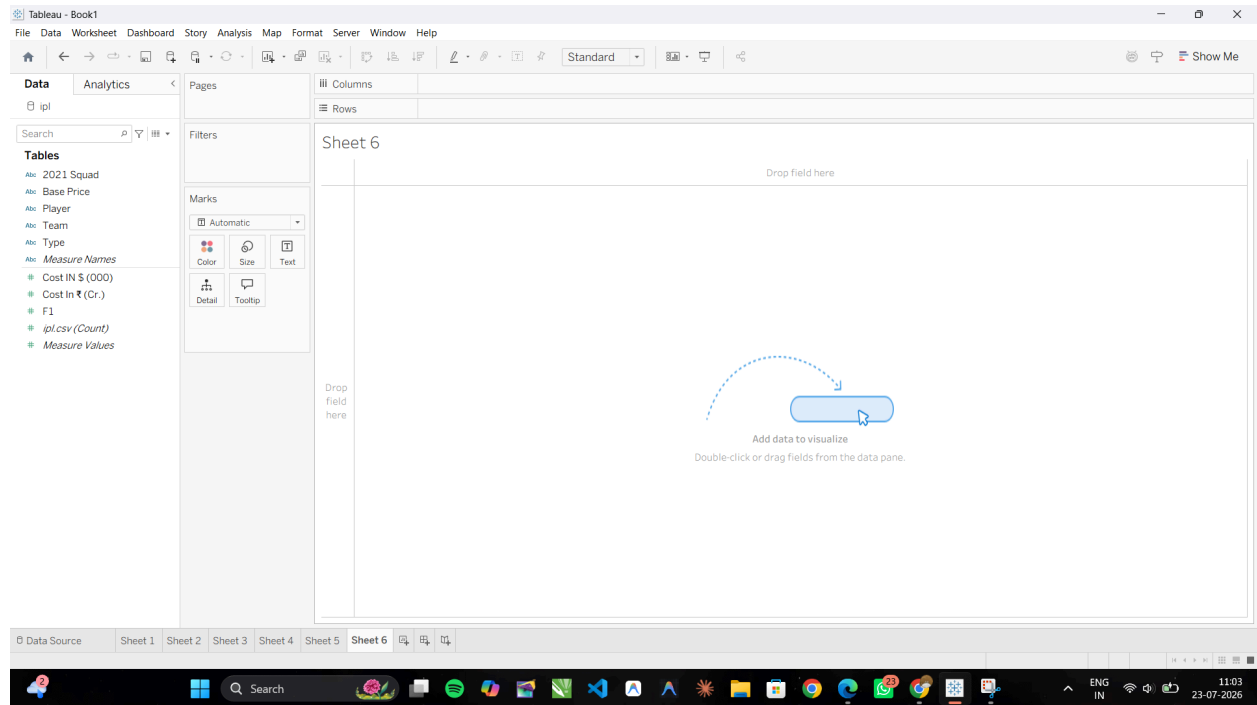
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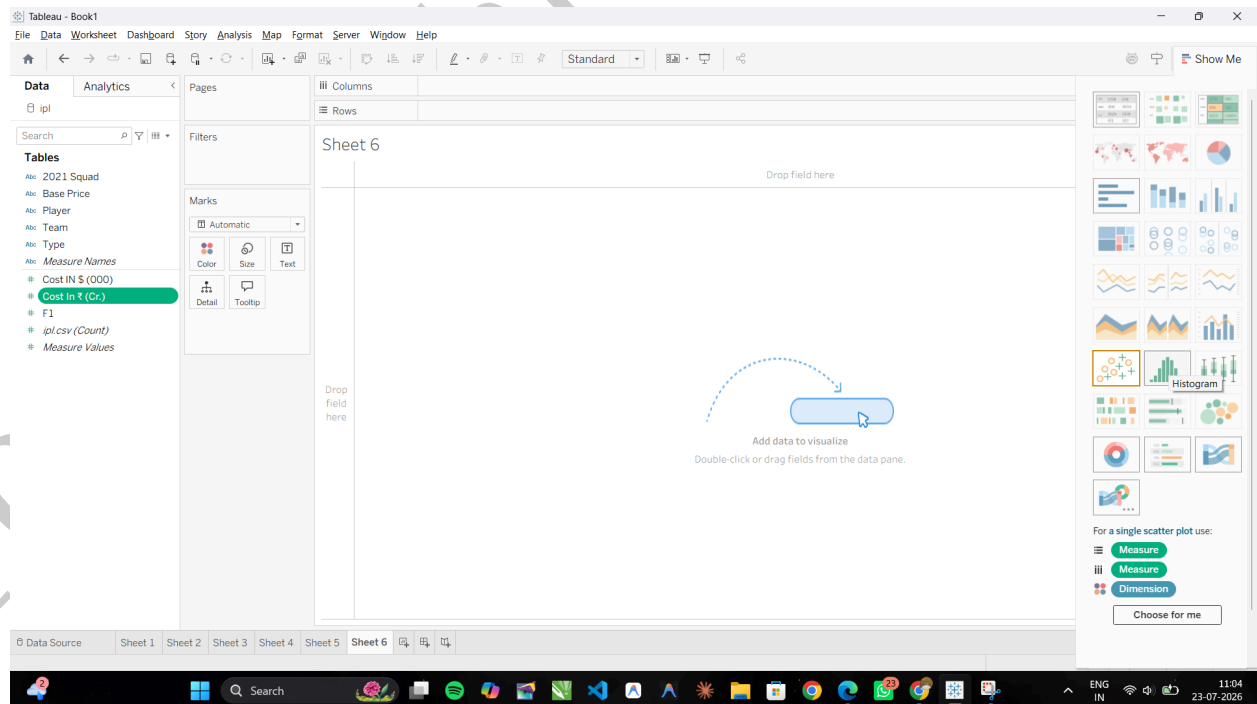
Step E: Developing Histograms for Distribution Analysis

1. Understanding Histograms in Tableau

Click on the New Worksheet icon to open a new worksheet (Sheet 6).



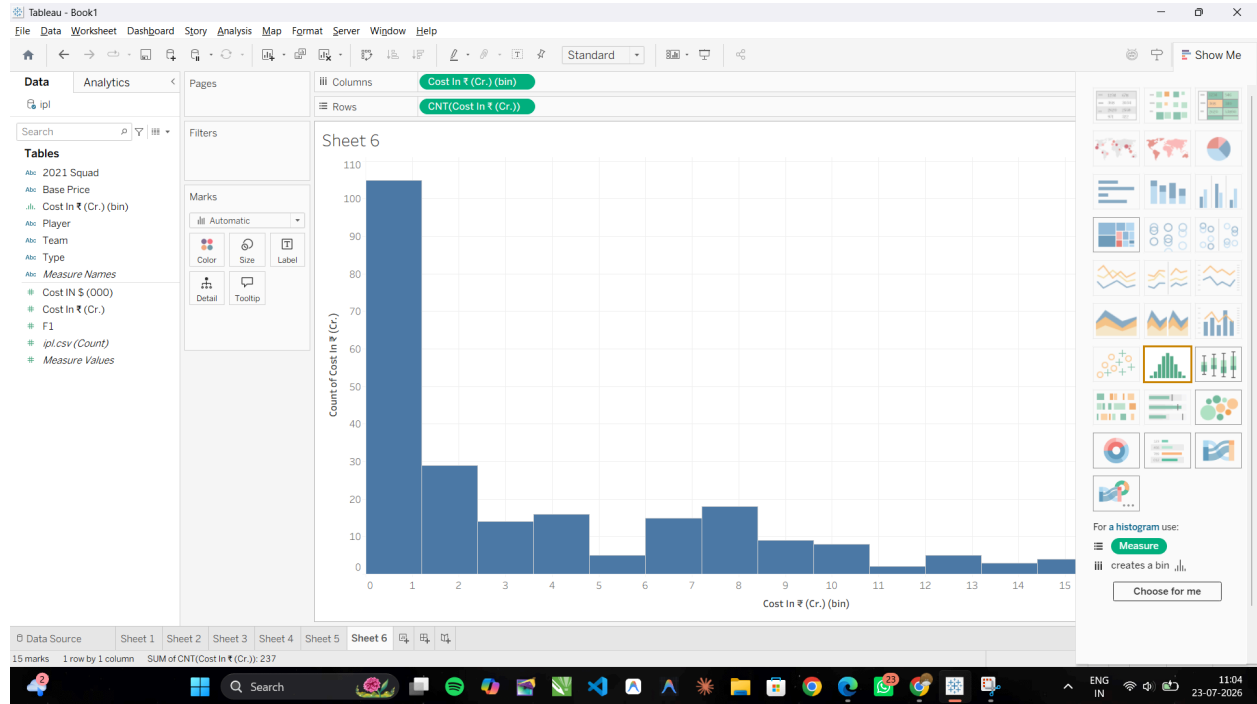
Drag **Cost In ₹ (Cr.)** from the **Data** pane to the **Columns** shelf.



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Right-click **Cost In ₹ (Cr.)** on the **Columns** shelf and select **Create** → **Bins**. Leave the default bin size and click **OK**.



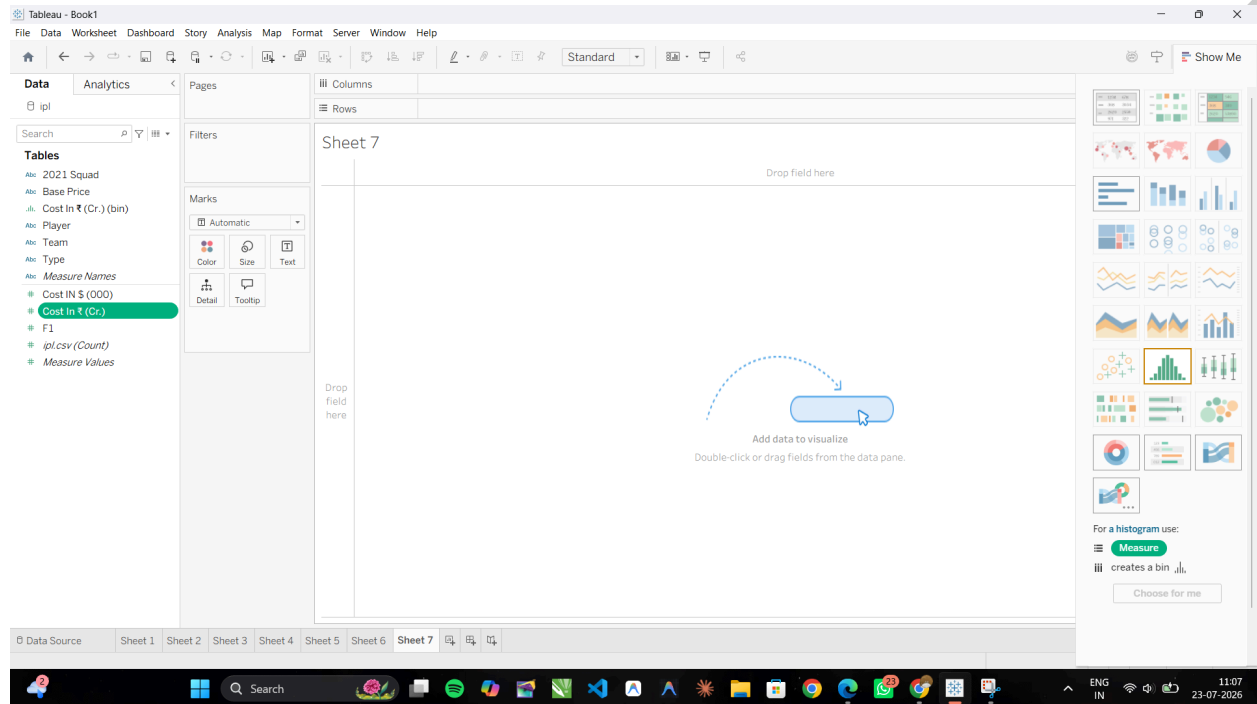
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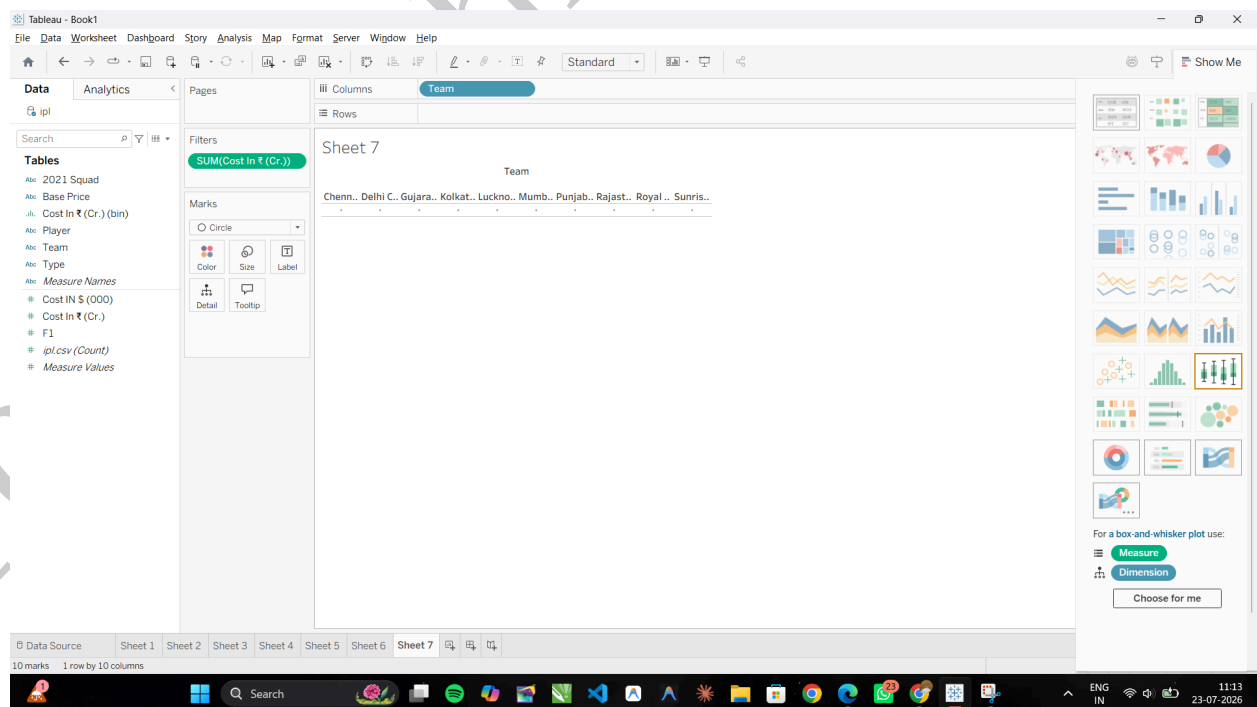
Step F: Building Box-and-Whisker Plots

1. Cleaning the View for Box Plots Building the Box-and-Whisker Plot

Click on the **New Worksheet** icon at the bottom of the screen to open a new worksheet (Sheet 7).



Drag **Team** to the **Columns** shelf and **Cost In ₹ (Cr.)** to the **Rows** shelf.



The screenshot displays the Tableau Desktop interface. The main view is a scatter plot on Sheet 7, titled 'Team'. The y-axis is labeled 'Cost in ₹ (Cr.)' and ranges from 0 to 90. The x-axis is labeled 'Team'. The plot shows data points for various teams, with a horizontal line at approximately 85. The interface includes a sidebar with filters and marks, a top menu bar, and a bottom status bar.

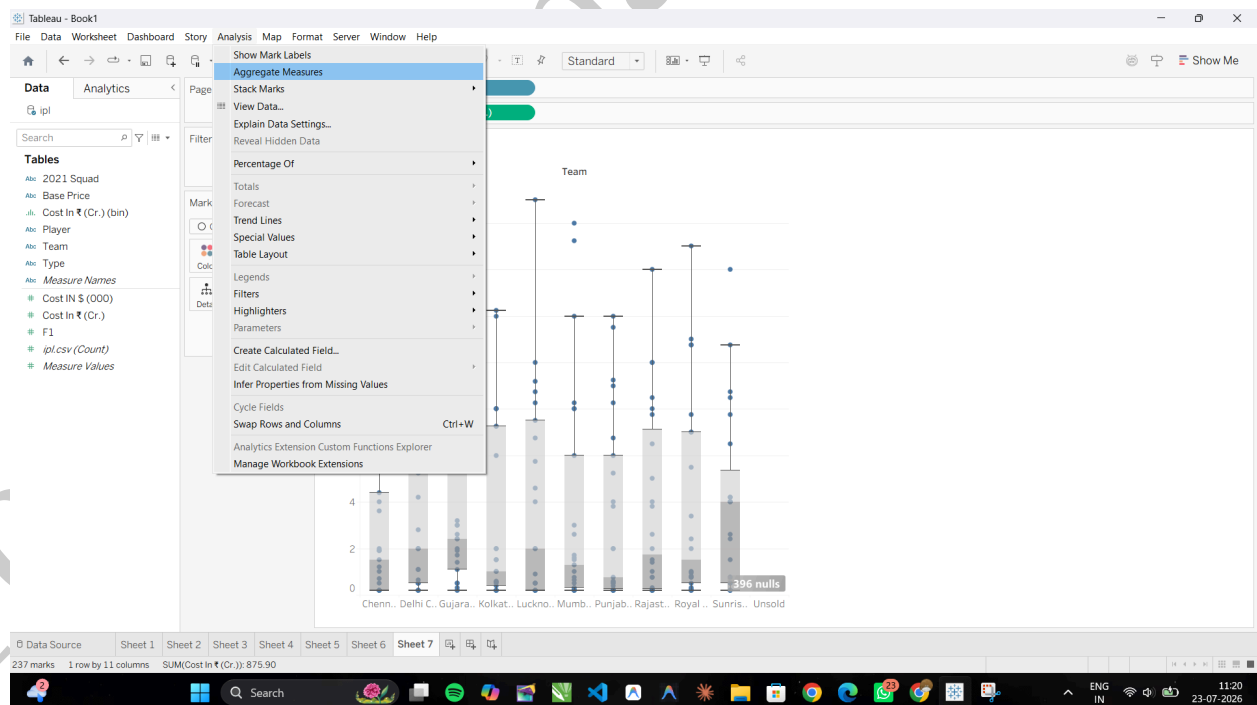
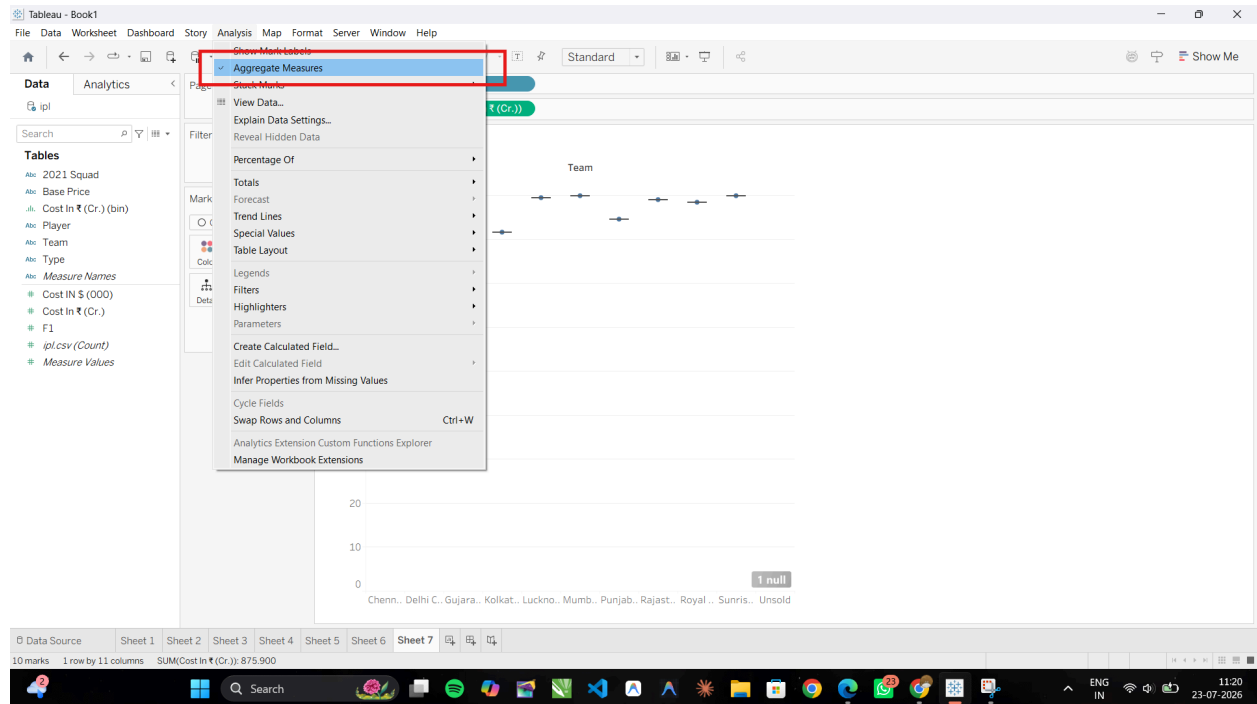
Tableau Desktop Interface Details:

- Top Menu Bar:** File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, Help.
- Top Toolbar:** Home, Back, Forward, Refresh, Download, Print, Copy, Paste, Undo, Redo, Erase, Lasso Select, Pan, Zoom In, Zoom Out, Standard, Show Me.
- Left Sidebar:**
 - Data:** ipl
 - Analytics:** SUM(Cost In ₹ (Cr.))
 - Filters:** SUM(Cost In ₹ (Cr.))
 - Marks:** Circle
 - Color:** Detail
 - Size:** Tooltip
 - Label:** Label
- Right Sidebar:** Show Me (various chart types)
- Bottom Status Bar:** 0 Data Source, Sheet 1, Sheet 2, Sheet 3, Sheet 4, Sheet 5, Sheet 6, Sheet 7, 10 marks, 1 row by 10 columns, SUM(Cost In ₹ (Cr.)): 875,900

The screenshot displays the Tableau Desktop interface with the 'Analysis' menu highlighted in the top navigation bar. The main view shows a dot plot visualization titled 'Sheet 7'. The Y-axis is labeled 'Cost In ₹ (Cr.)' and ranges from 0 to 90. The X-axis is labeled 'Team' and lists various teams: Chenn., Delhi C., Gujara., Kolkat., Luckno., Mumb., Punjab., Rajast., Royal., Sunris., and Unsold. The data points are represented by dots with horizontal error bars. The status bar at the bottom indicates '10 marks' and '1 row by 11 columns'.

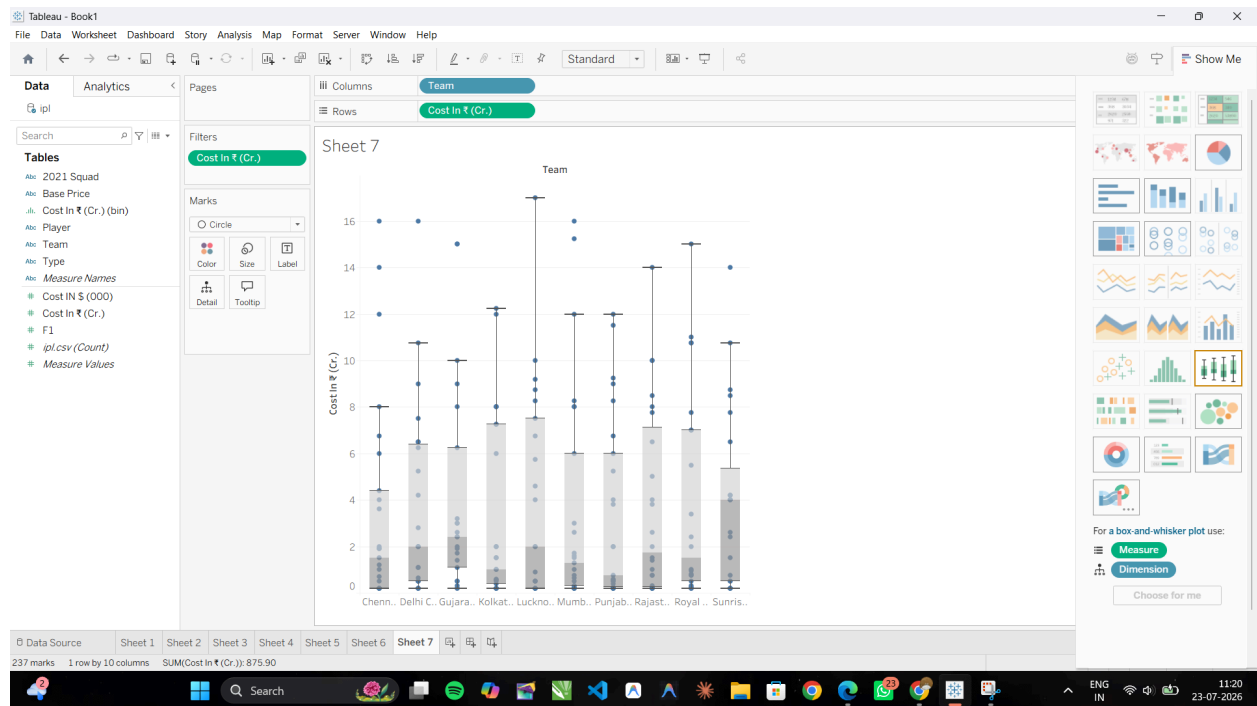
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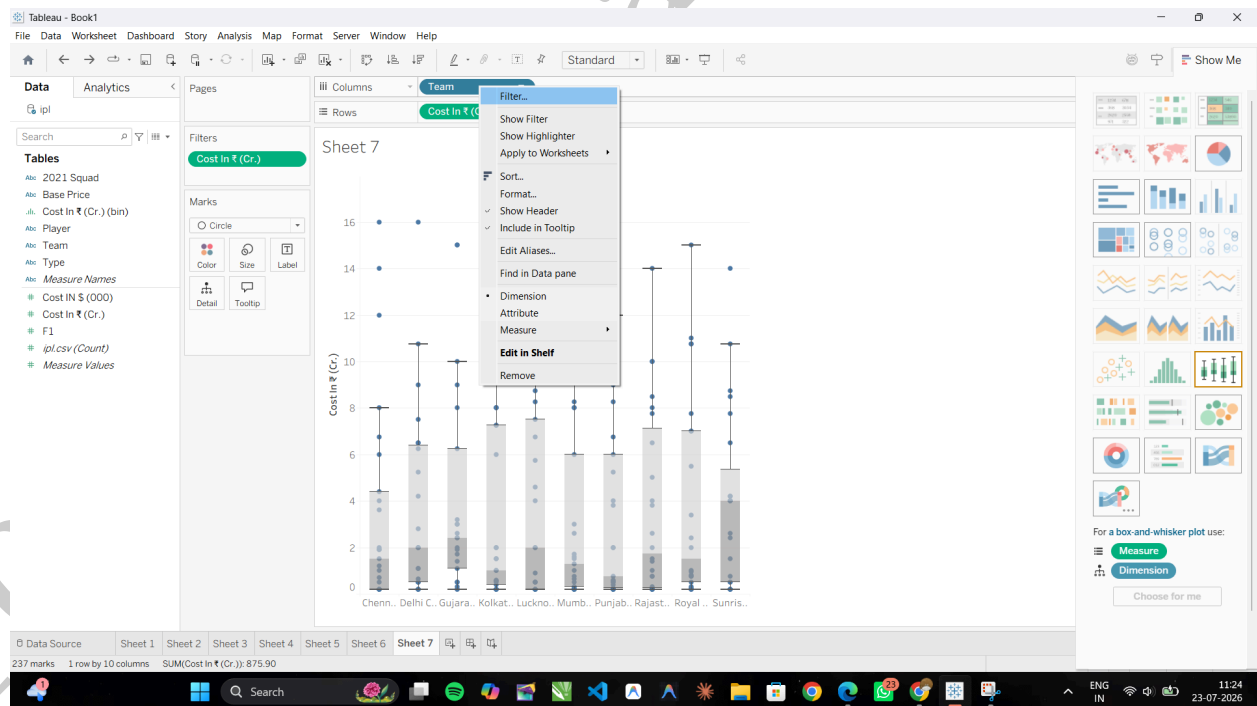


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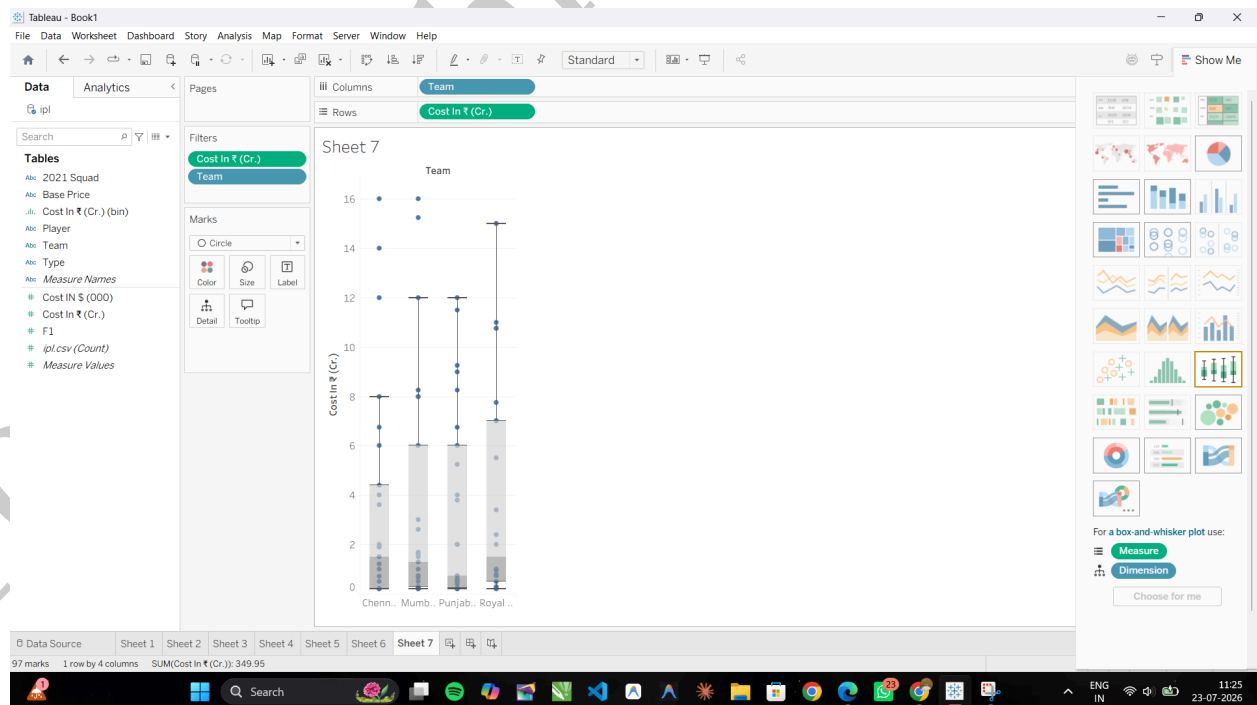
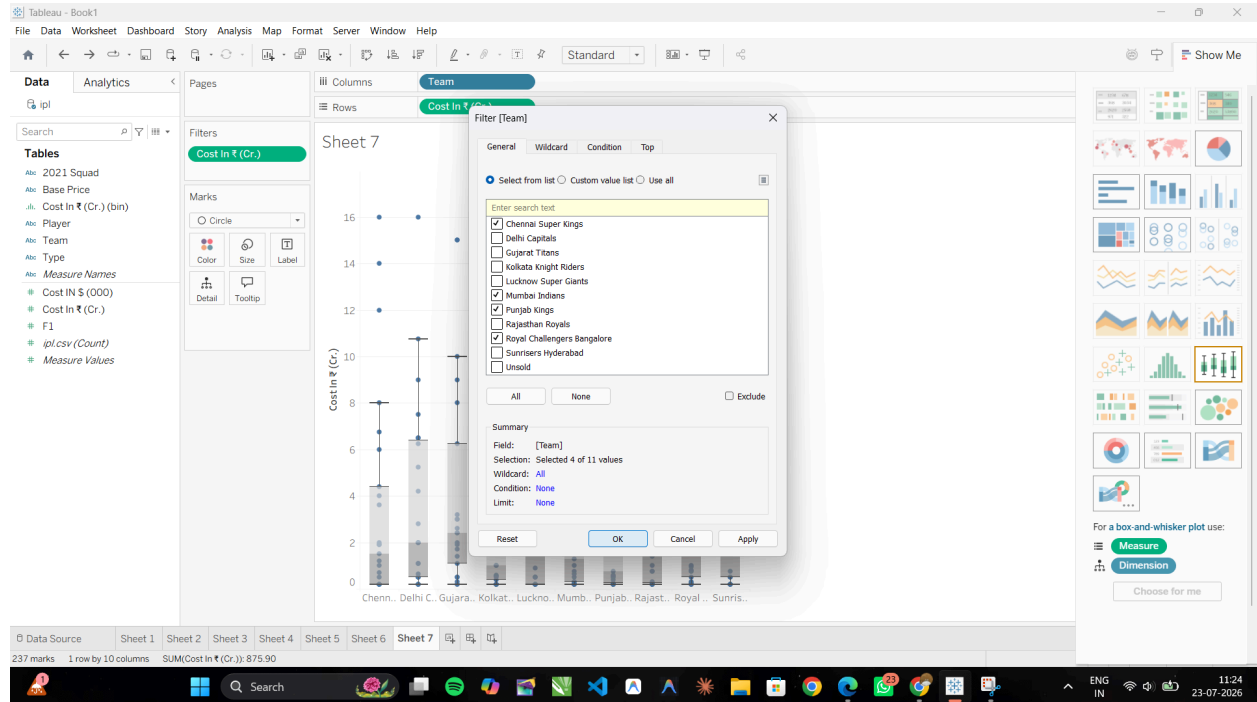
Filter: Right click on Teams and select Filters



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Select any Teams and Click OK



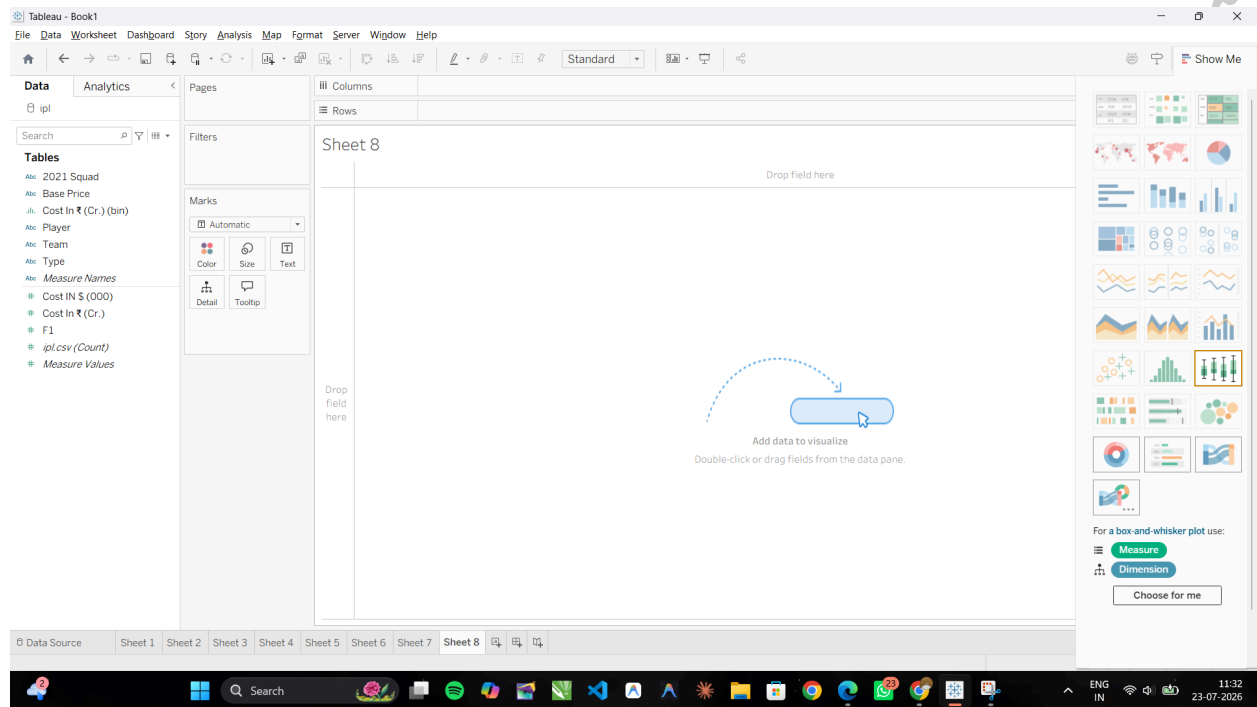
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Step G: Using Gantt Charts and Advanced Date Visualizations

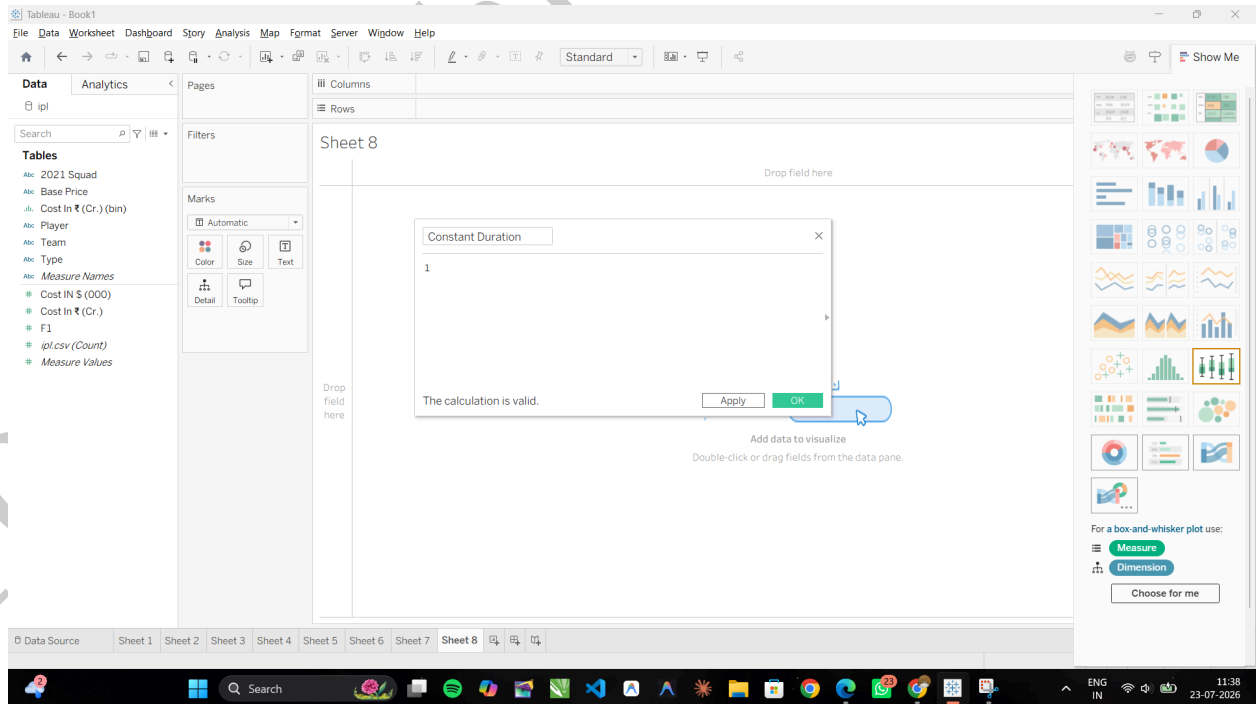
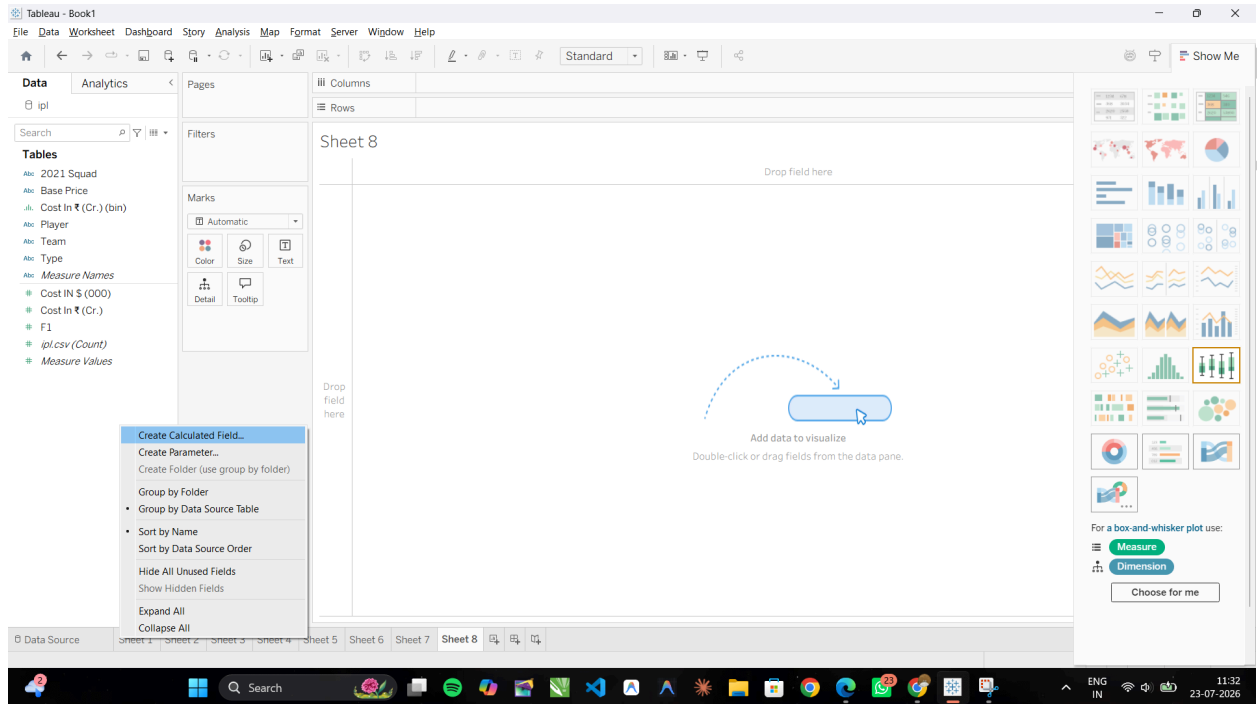
1. Understanding Gantt Charts in Tableau Building an Advanced Date Gantt Timeline

Click on the New Worksheet icon at the bottom of the screen to open a new worksheet (Sheet 8).



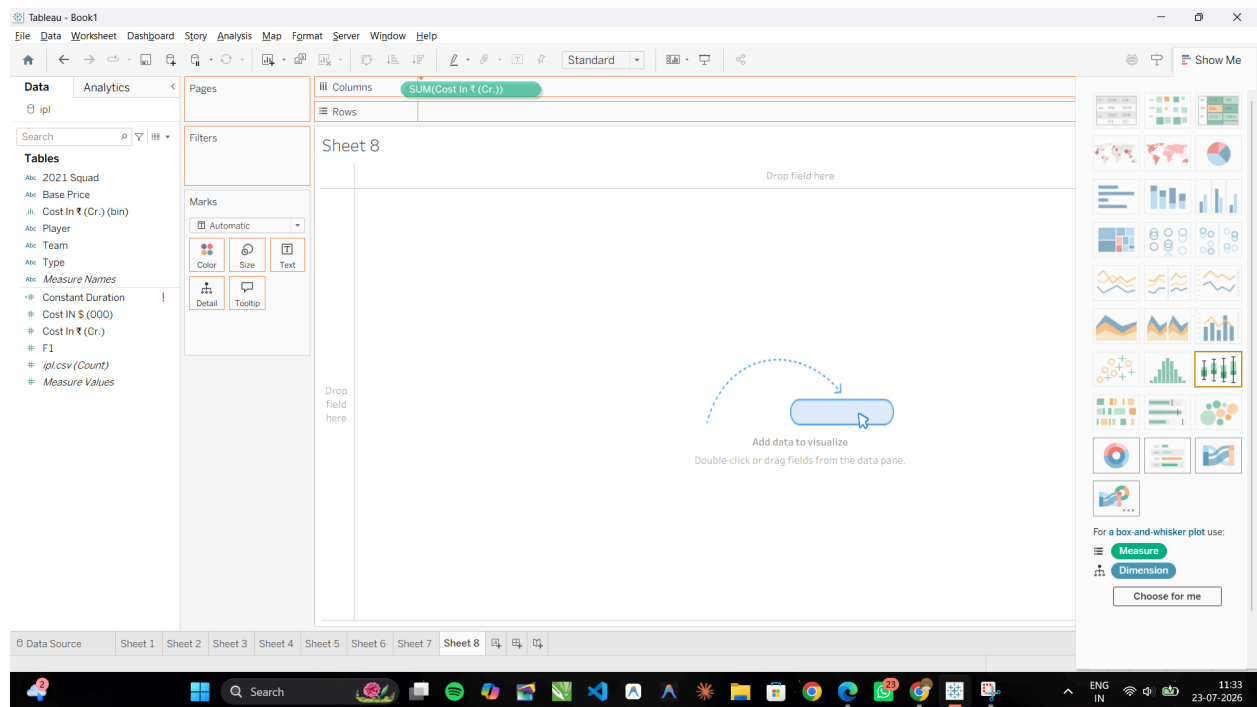
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Create a calculated field by selecting **Analysis** → **Create Calculated Field**, name it **Constant Duration**, enter the formula **1**, and click **OK**.

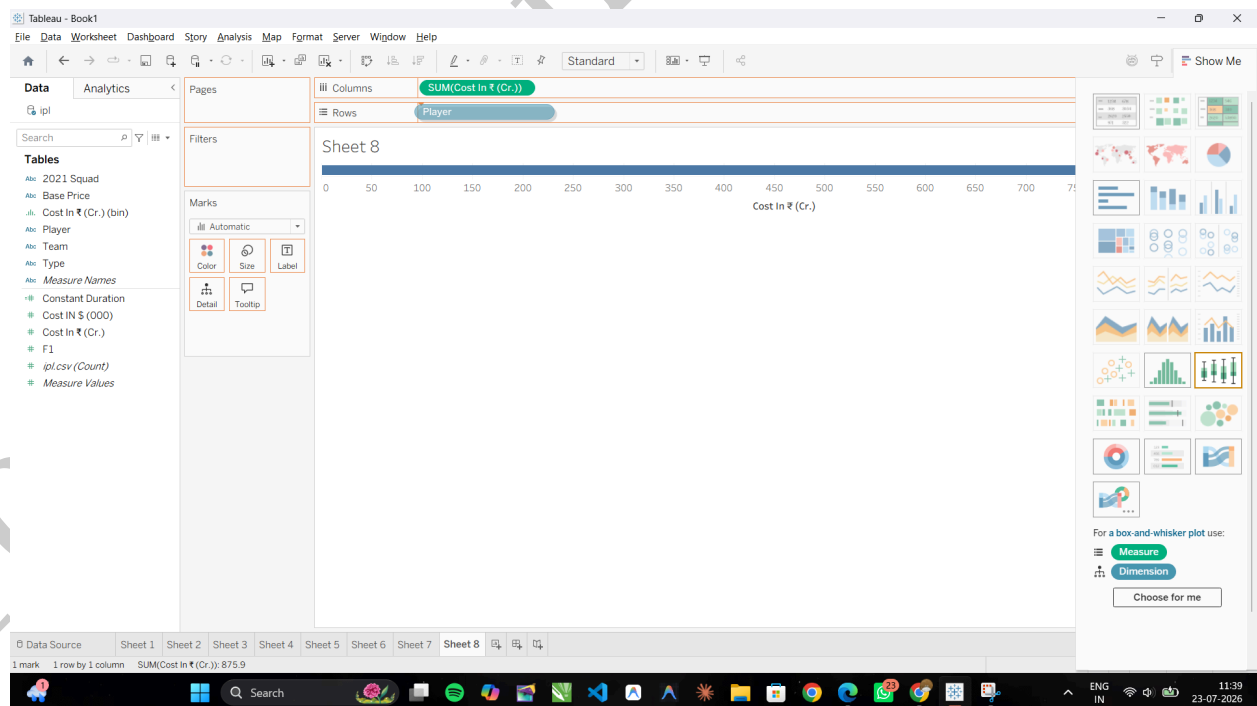


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Drag Cost In ₹ (Cr.) to the Columns shelf.



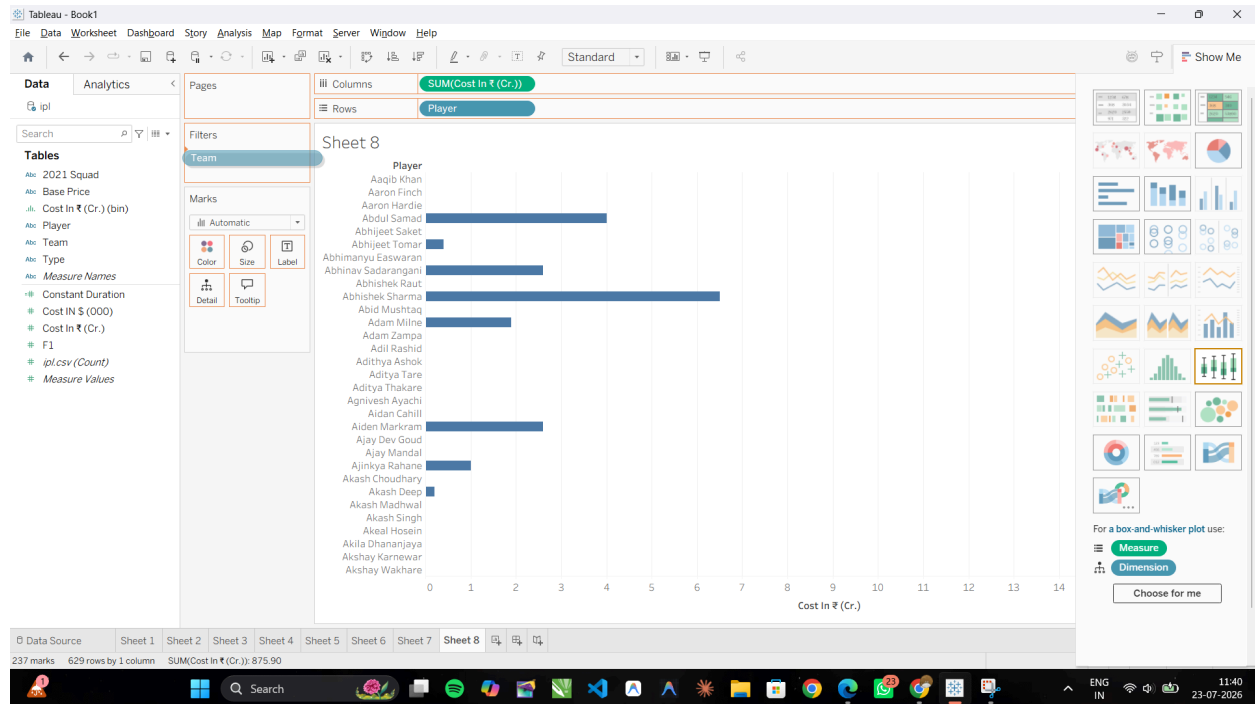
Drag Player to the Rows shelf.



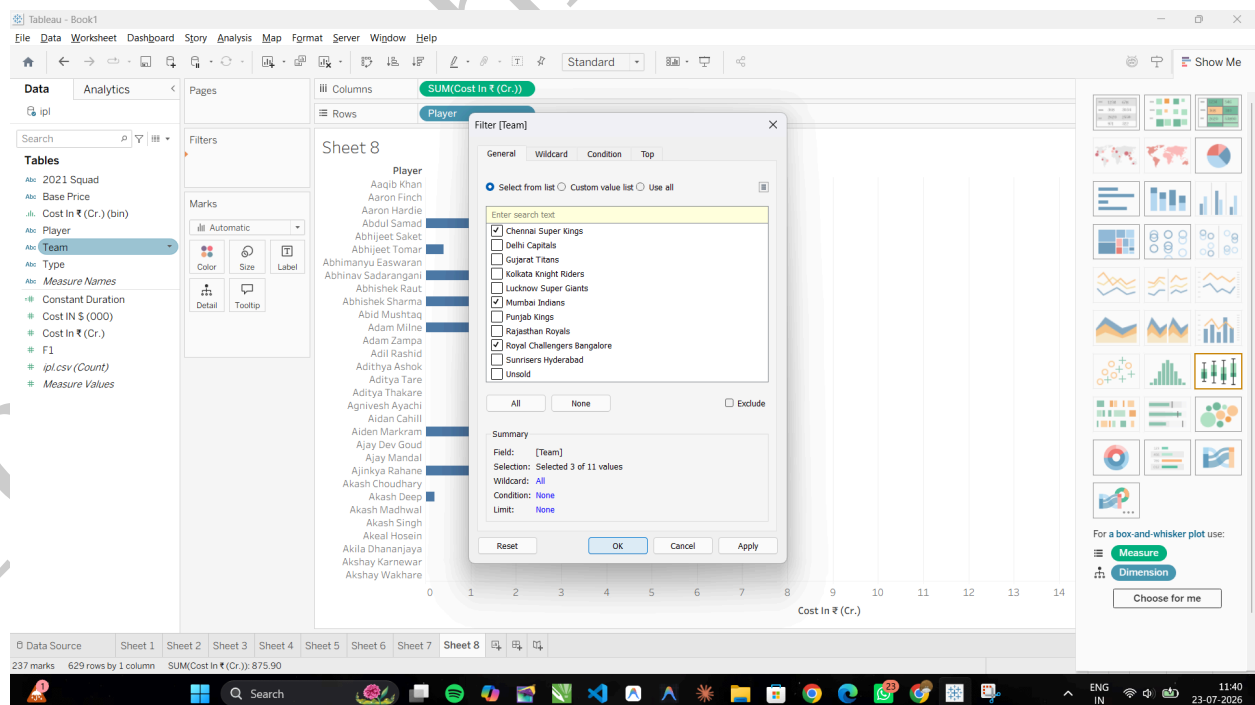
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Drag Teams to the Size option in the Marks card.



Select any Team

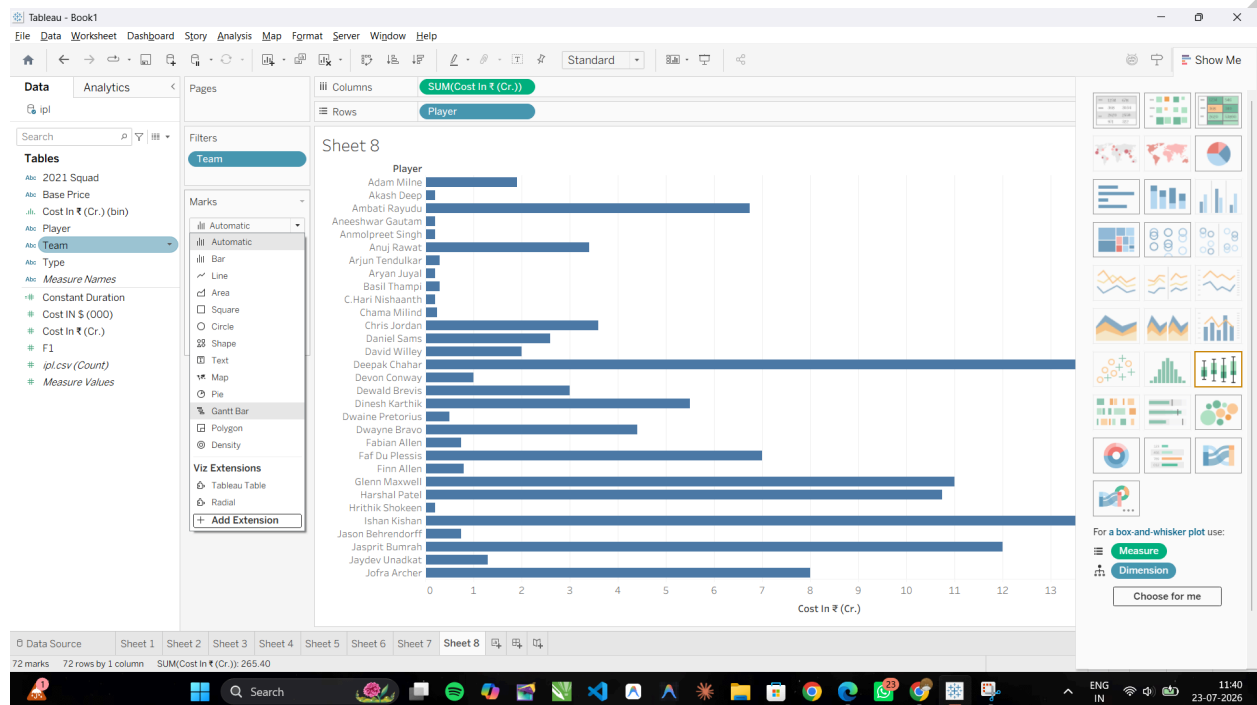


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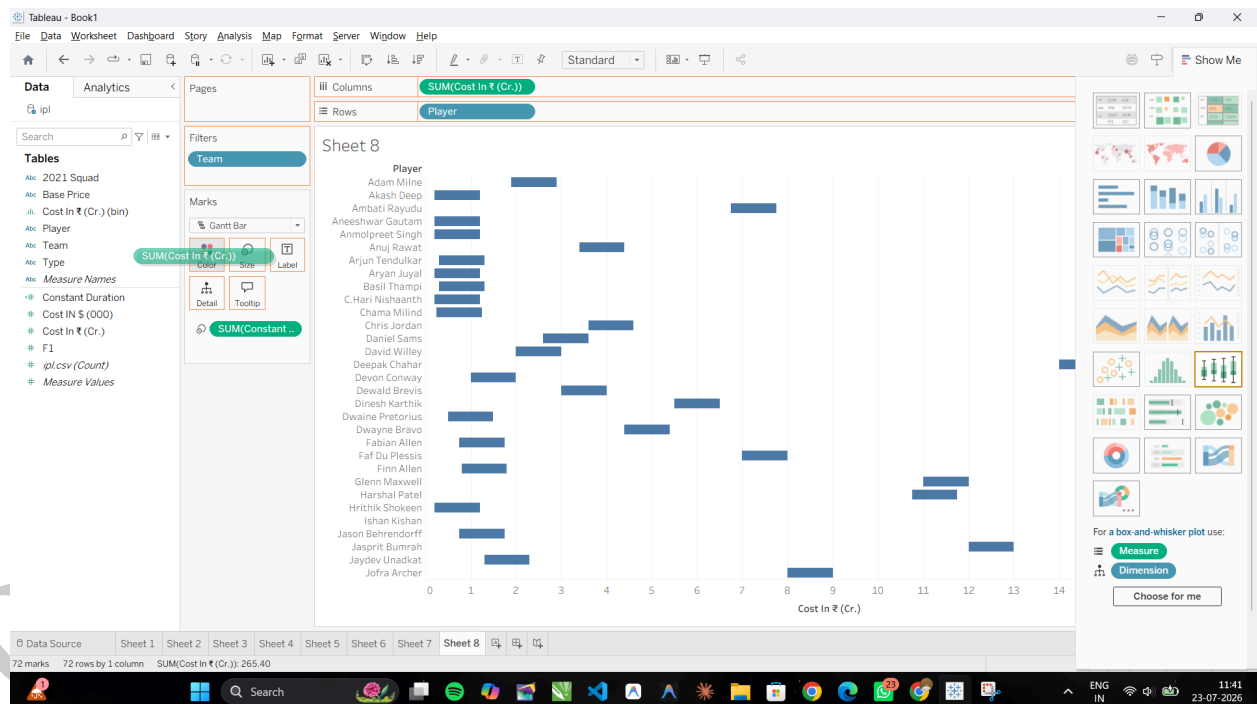
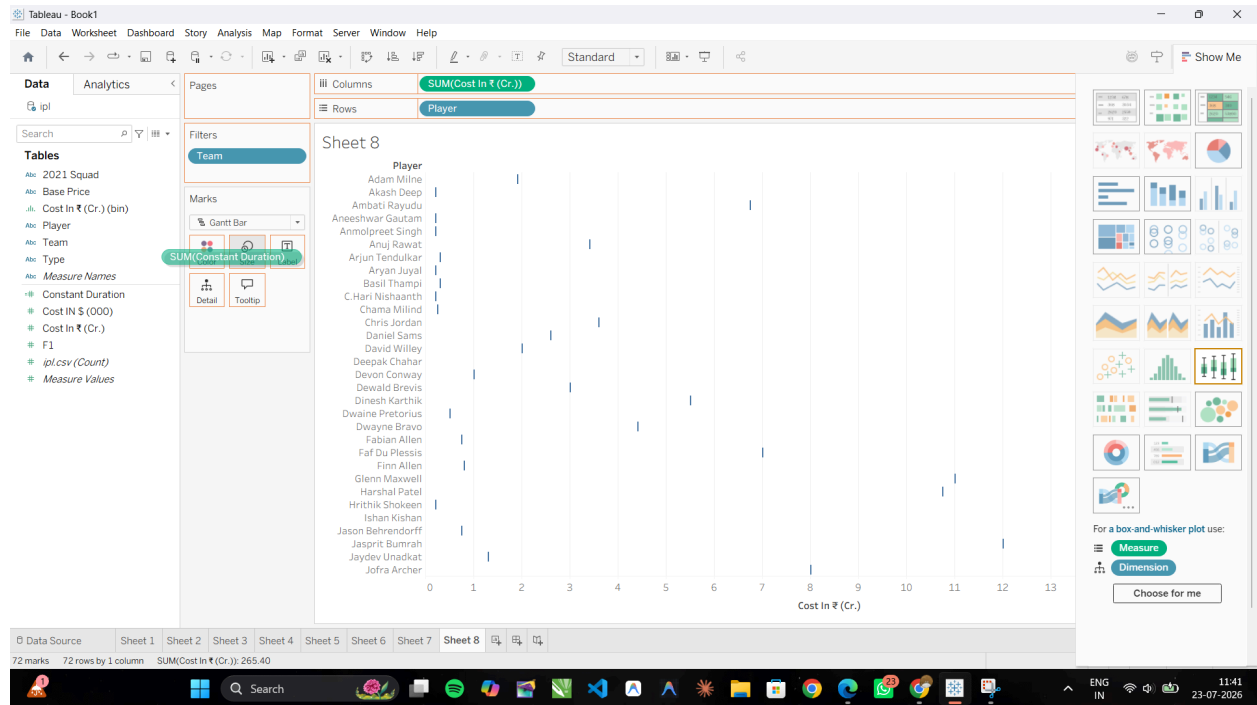
Click on Automatic and Select Gantt Bar



Drag **Constant Duration** to the **Size** option in the **Marks** card.

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